Tecnologie del Linguaggio Naturale

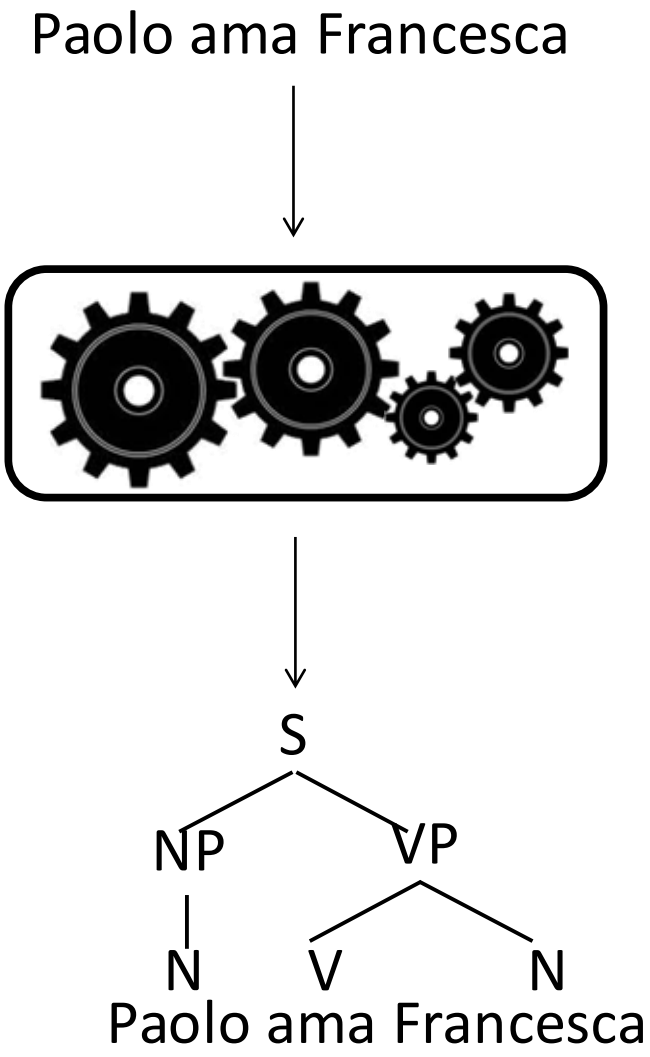
Parte Prima

Lezione n. 04-2

> Sintassi: la performance a costituenti

2 Marzo, 2026

Parser



Outline

- Parser anatomy
- Top-down vs. bottom-up
- CKY algorithm: Cocke-Kasami-Younger
- Parsing probabilistico e TB grammars
- Parsing parziale

Outline

- **Parser anatomy**
- Top-down vs. bottom-up
- CKY algorithm: Cocke-Kasami-Younger
- Parsing probabilistico e TB grammars
- Parsing parziale

Parser anatomy (Steedman)

(1) Grammar (or automa model)

Context-Free, TAG, CCG, Dependency …

(2) Algorithm

I. Search strategy

top-down, bottom-up, left-to-right, …

II. Memory organization

back-tracking, dynamic programming, …

(3) Oracle

Probabilistic, rule-based, …

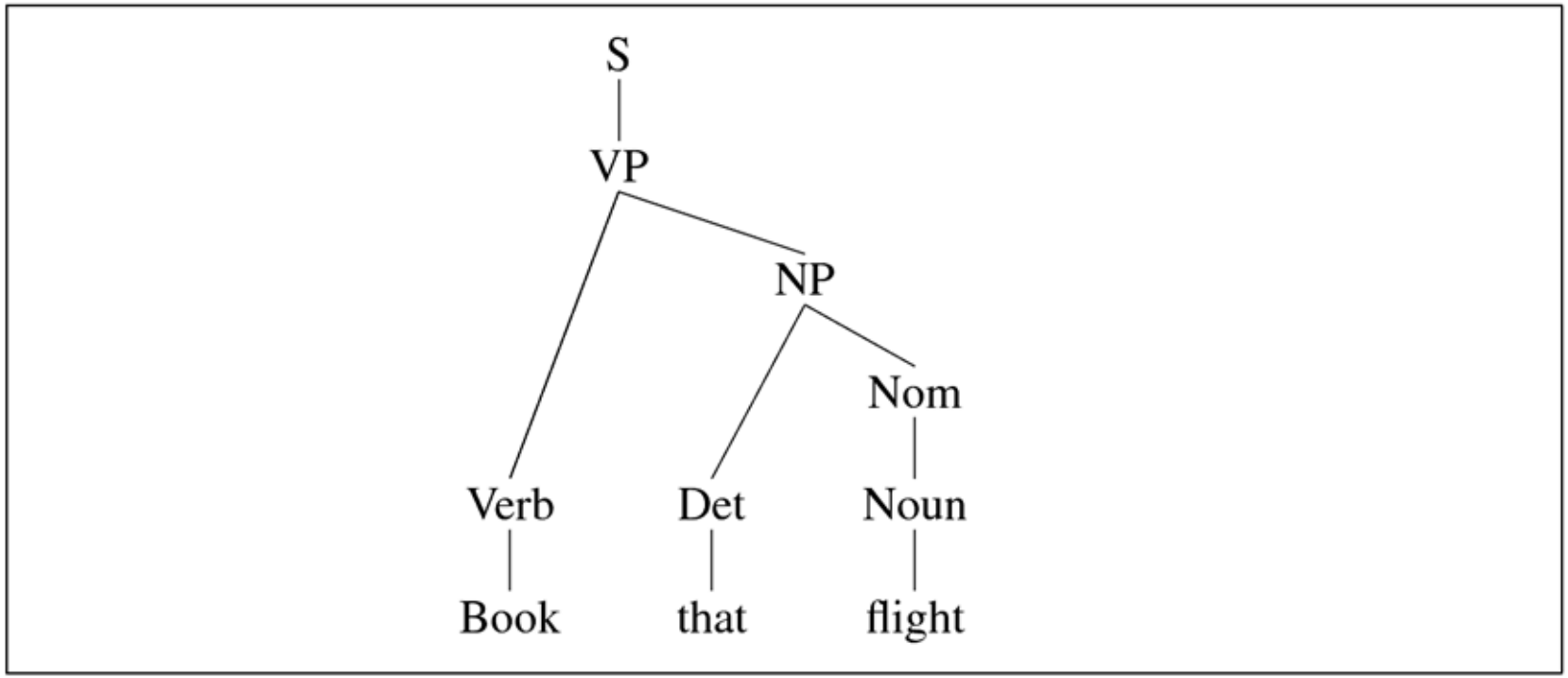
Outline

- Parser anatomy
- **Top-down vs. bottom-up**
- CKY algorithm: Cocke-Kasami-Younger
- Parsing probabilistico e TB grammars
- Parsing parziale

Grammar

$S \rightarrow NP\ VP$	$Det \rightarrow that \mid this \mid a$
$S \rightarrow Aux\ NP\ VP$	$Noun \rightarrow book \mid flight \mid meal \mid money$
$S \rightarrow VP$	$Verb \rightarrow book \mid include \mid prefer$
$NP \rightarrow Det\ Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	
$Nominal \rightarrow Noun\ Nominal$	$Prep \rightarrow from \mid to \mid on$
$NP \rightarrow Proper\text{-}Noun$	$Proper\text{-}Noun \rightarrow Houston \mid TWA$
$VP \rightarrow Verb$	
$VP \rightarrow Verb\ NP$	$Nominal \rightarrow Nominal\ PP$

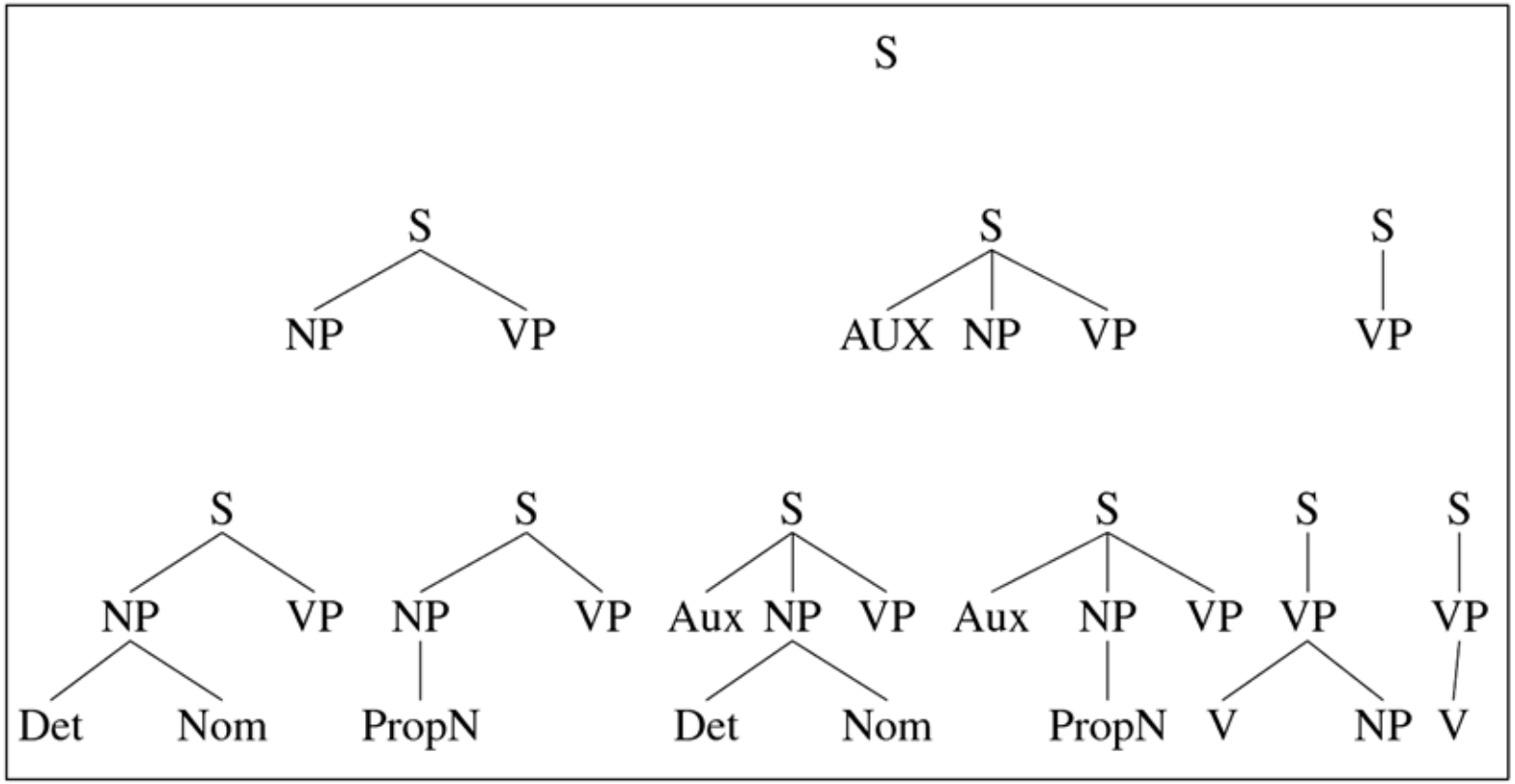
Target Parse



Information sources

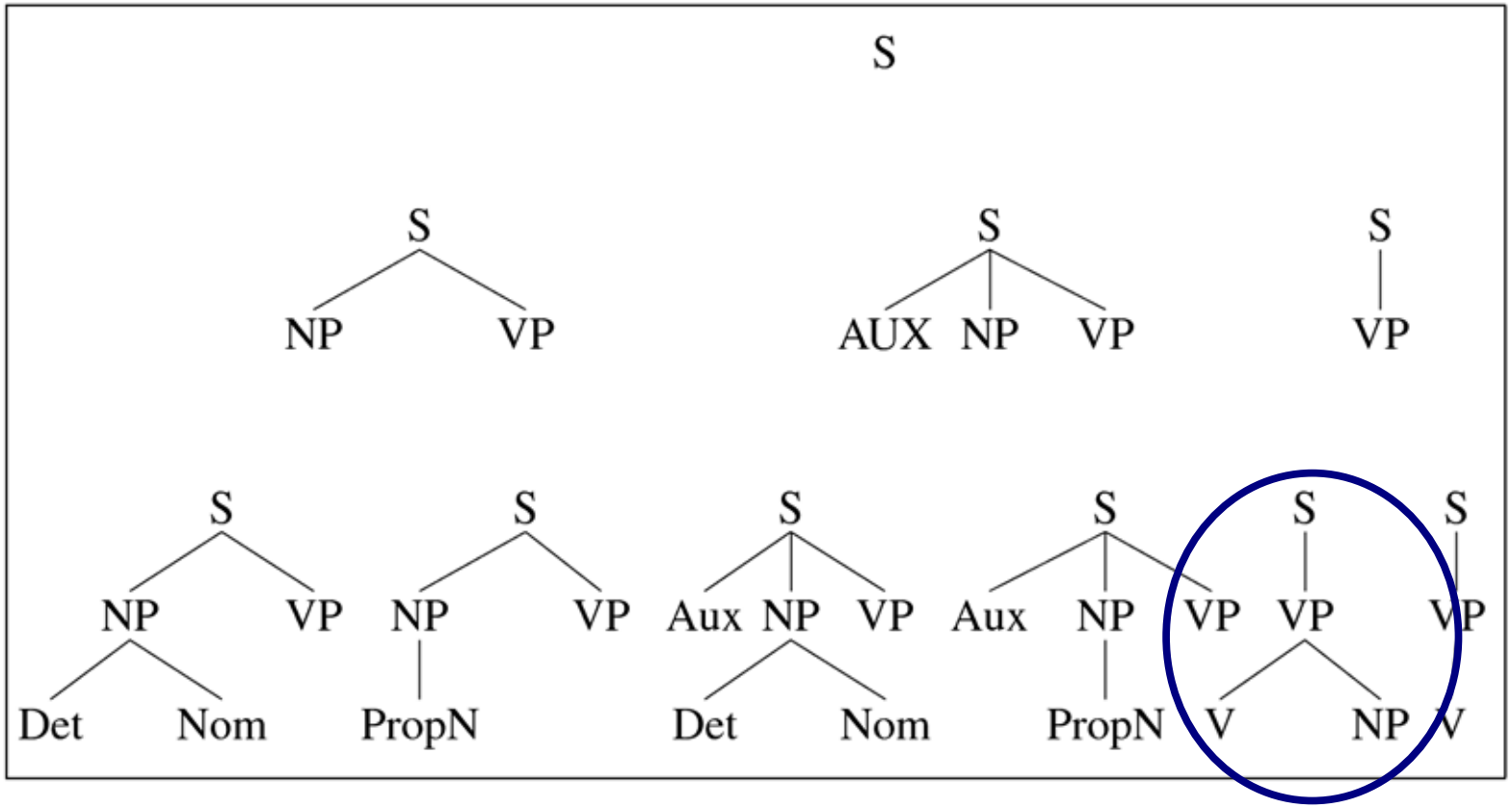
- The grammar -> **goal**-directed parsing -> top-down
  - Solo ricerche che possono portare a risposte corrette, cioè con radice S
  - Ma comporta la creazione di alberi non compatibili con le parole
  - Razionalisti
- The sentence -> **data**-directed parsing -> bottom-up
  - Solo ricerche compatabili con le parole
  - Ma comporta la creazione di alberi non corretti
  - Empiricisti

Top-Down



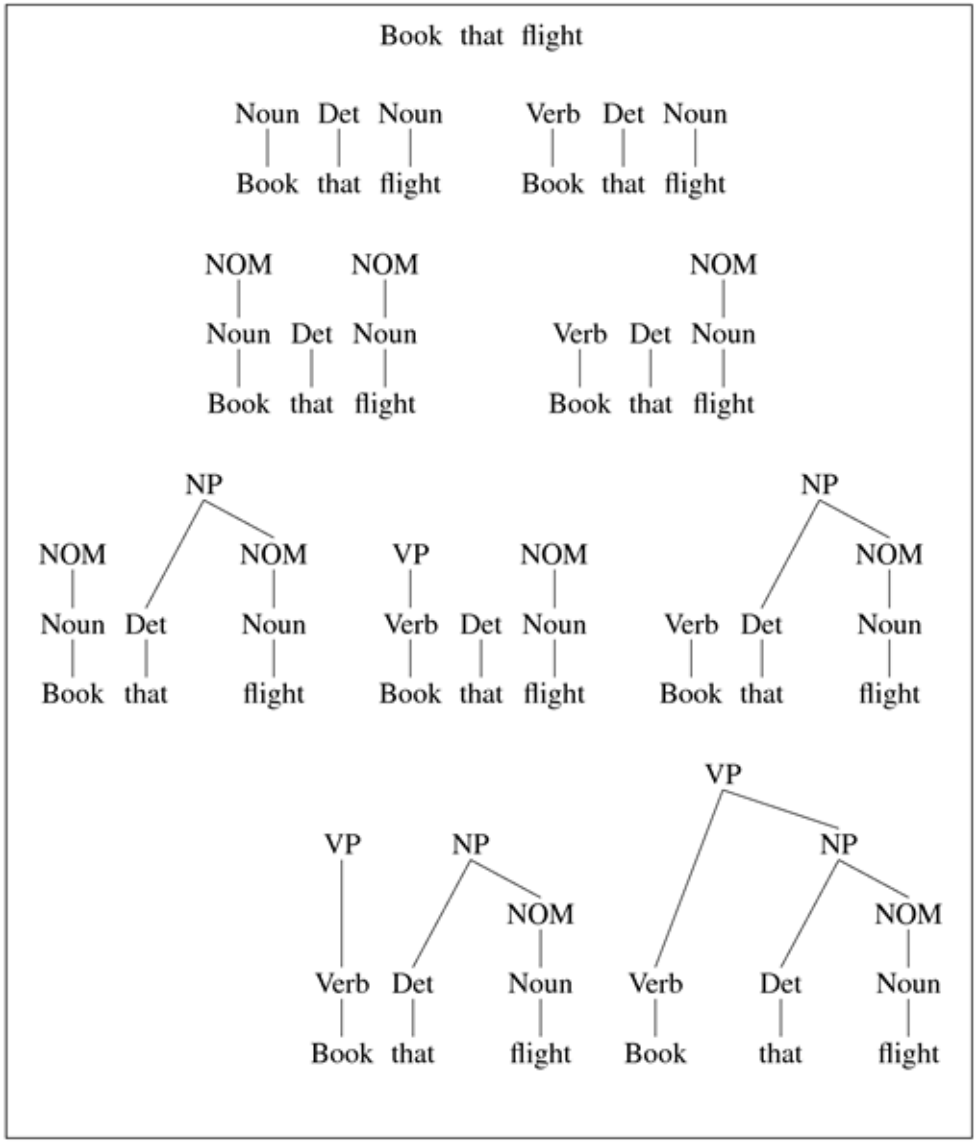
Book that flight

Top-Down

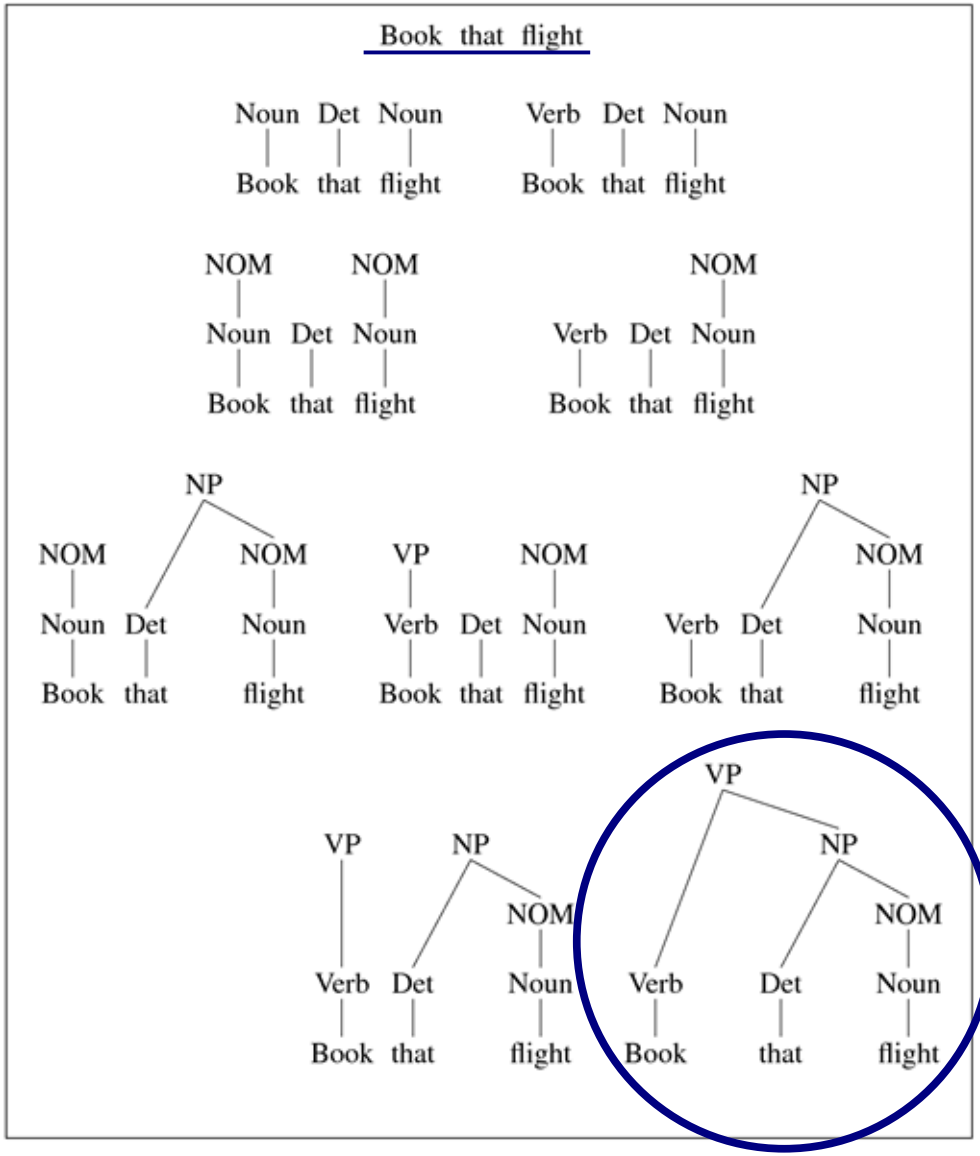


Book that flight

Bottom-Up


Book that flight
Noun Det Noun
Book that flight
Verb Det Noun
Book that flight
NOM NOM
Noun Det Noun
Book that flight
NOM
Verb Det Noun
Book that flight
NP
NOM NOM
Noun Det Noun
Book that flight
VP NOM
Verb Det Noun
Book that flight
NP
NOM
Verb Det Noun
Book that flight
VP
VP NP
NOM
Verb Det Noun
Book that flight
VP
NP
NP
NOM
Verb Det Noun
Book that flight

Bottom-Up


Book that flight
Noun Det Noun
Book that flight
Verb Det Noun
Book that flight
NOM NOM
Noun Det Noun
Book that flight
NOM
Verb Det Noun
Book that flight
NP
NOM NOM
Noun Det Noun
Book that flight
VP NOM
Verb Det Noun
Book that flight
NP
NOM
Verb Det Noun
Book that flight
VP
VP NP
NOM
Verb Det Noun
Book that flight
VP
NP NP
NOM NOM
Verb Det Noun
Book that flight

Parser A

(1) Grammar

Context-Free, ...

(2) Algorithm

I. Search strategy

top-down, bottom-up, left-to-right, ...

II. Memory organization

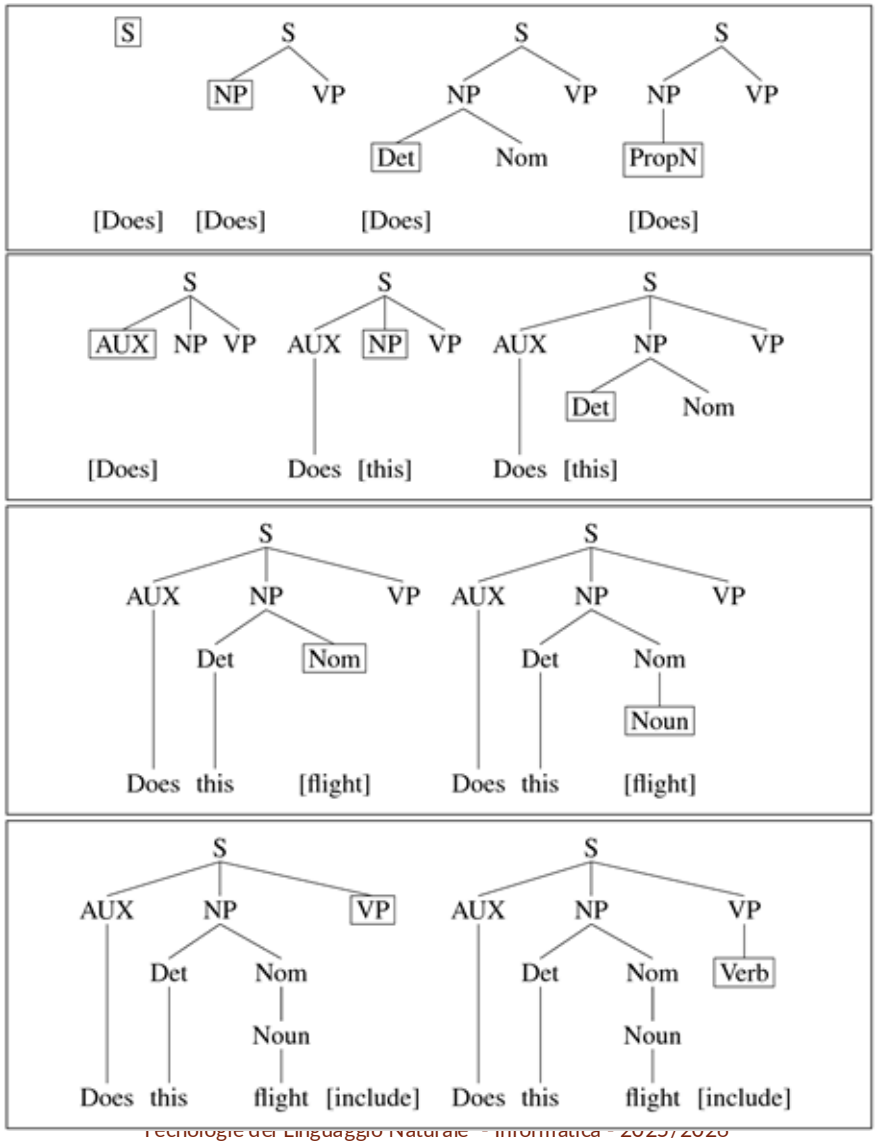
back-tracking, dynamic programming, ...

(3) Oracle

Probabilistic, rule-based, ...

Parser A - 1

Does this flight include a meal …



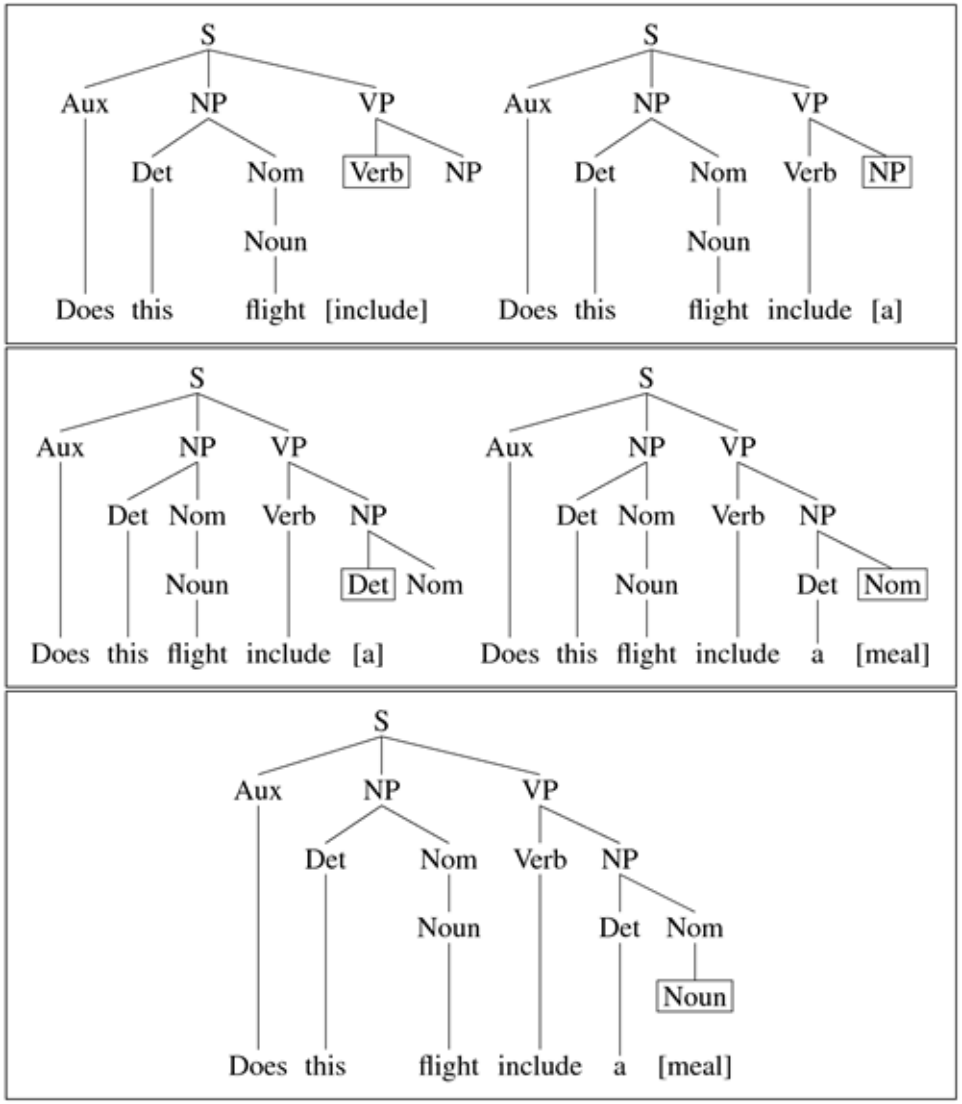
S→NP VP
NP→DET Nom
NP→Prop N

S→AUX NP VP
AUX→does
NP→DET Nom

DET→this
Nom→Noun

Noun→flight
VP→Verb

Parser A - 2



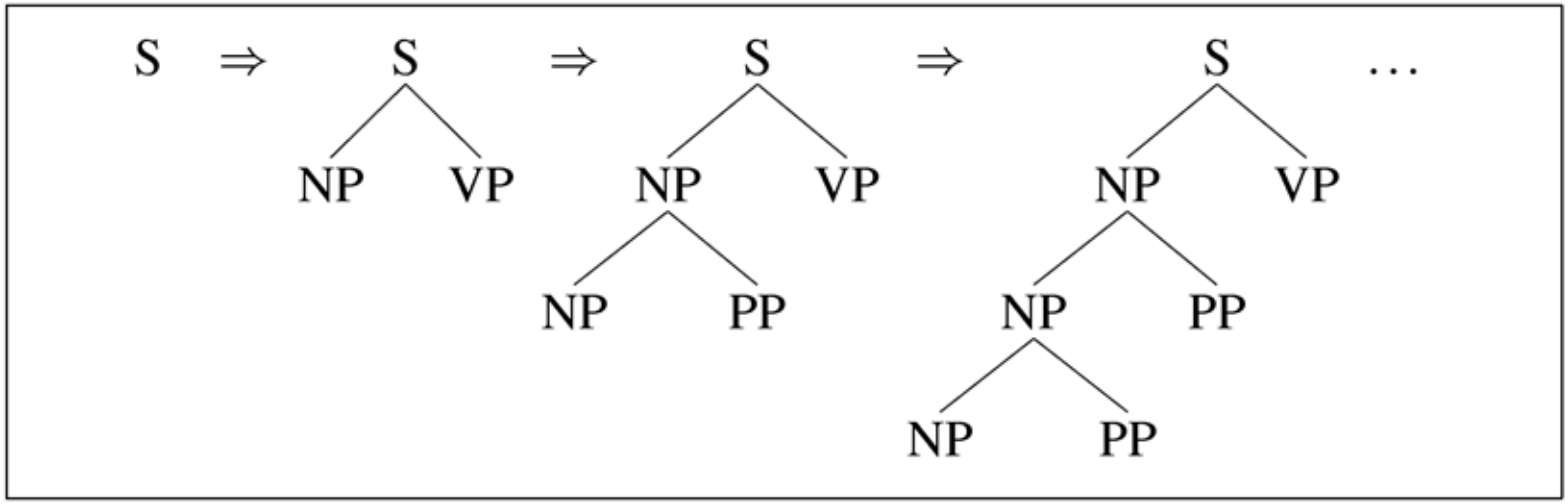
VP→Verb NP
Verb→include

NP→Det Nom
Det→a

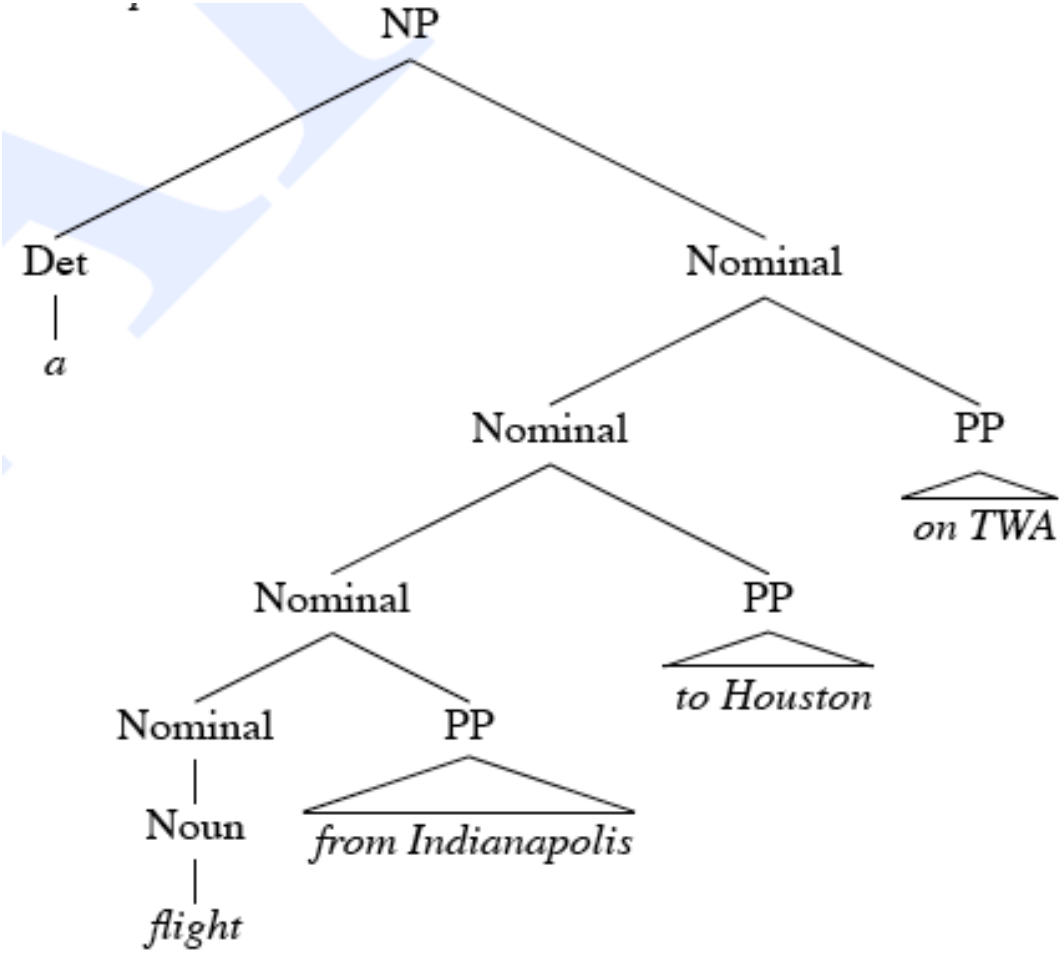
Nom→Noun
Noun→meal

Left-Recursion

$$NP \rightarrow NP\ PP$$



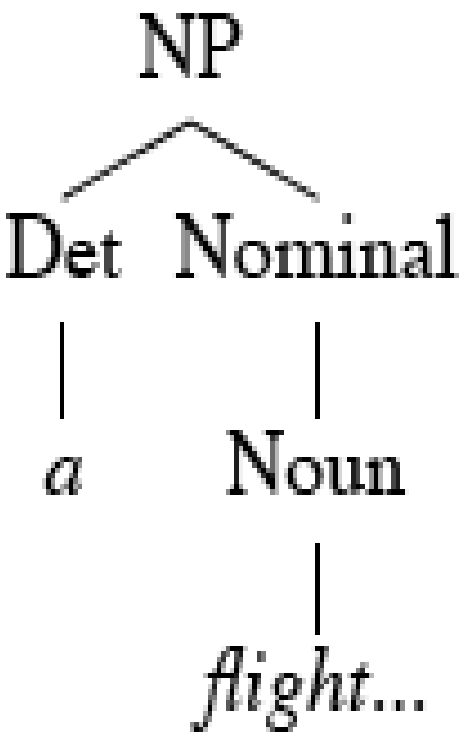
Subtree parsing -> Shared Sub-problems



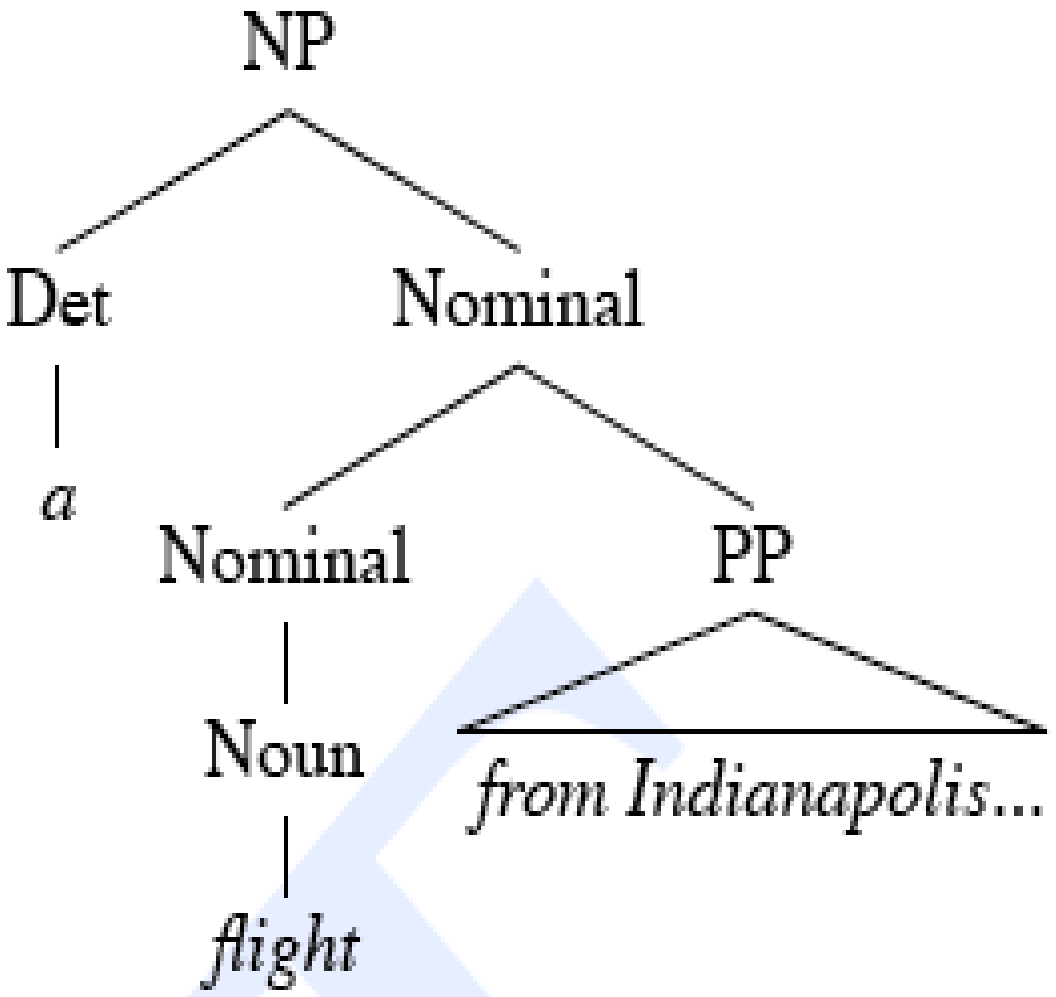
Shared Sub-problems

- Top-down parse che ha già espanso la NP
- 2 possibili continuazioni:
  - Nominal -> Noun
  - Nominal -> Nominal PP
- Se scelgo sempre la prima ottengo ...

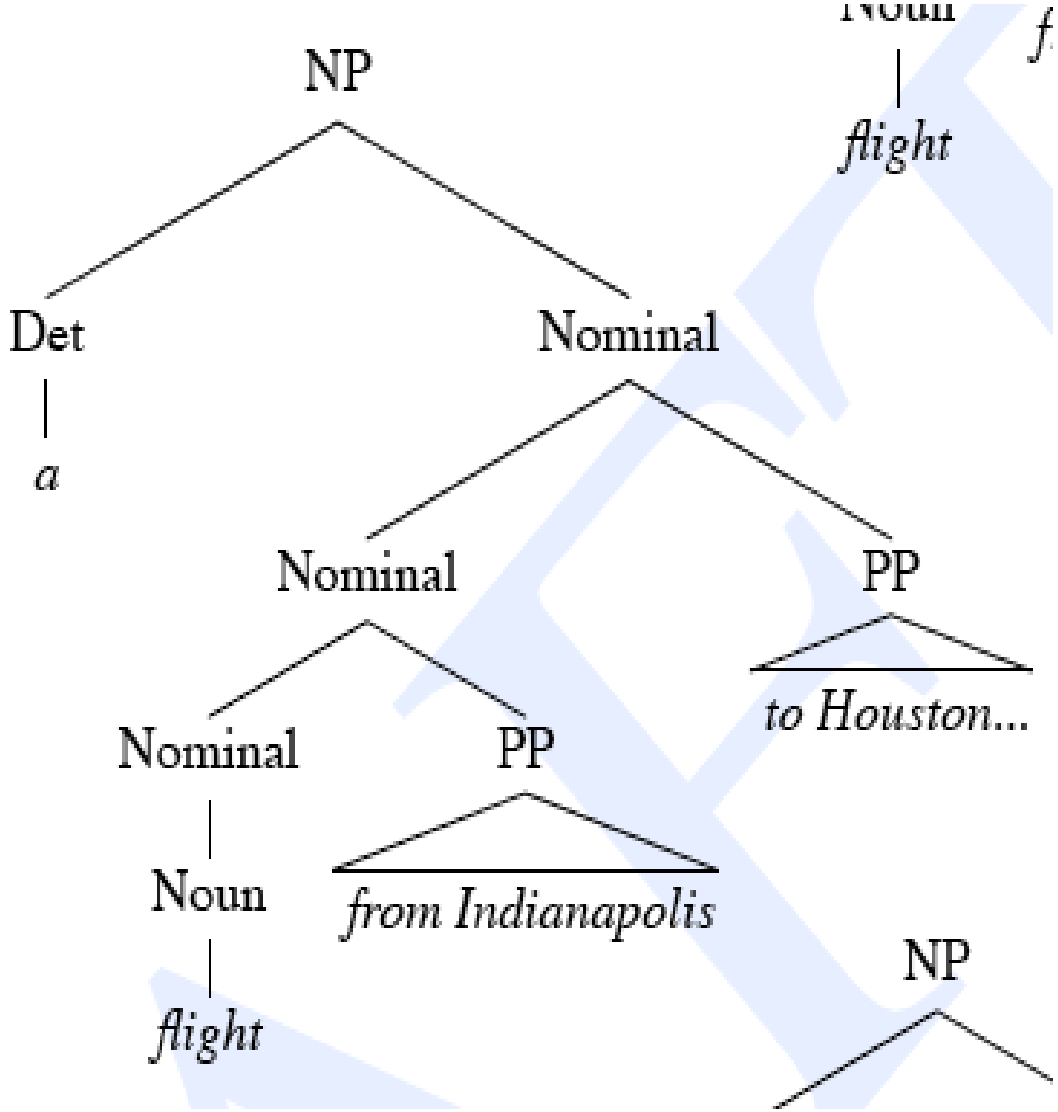
Shared Sub-problems



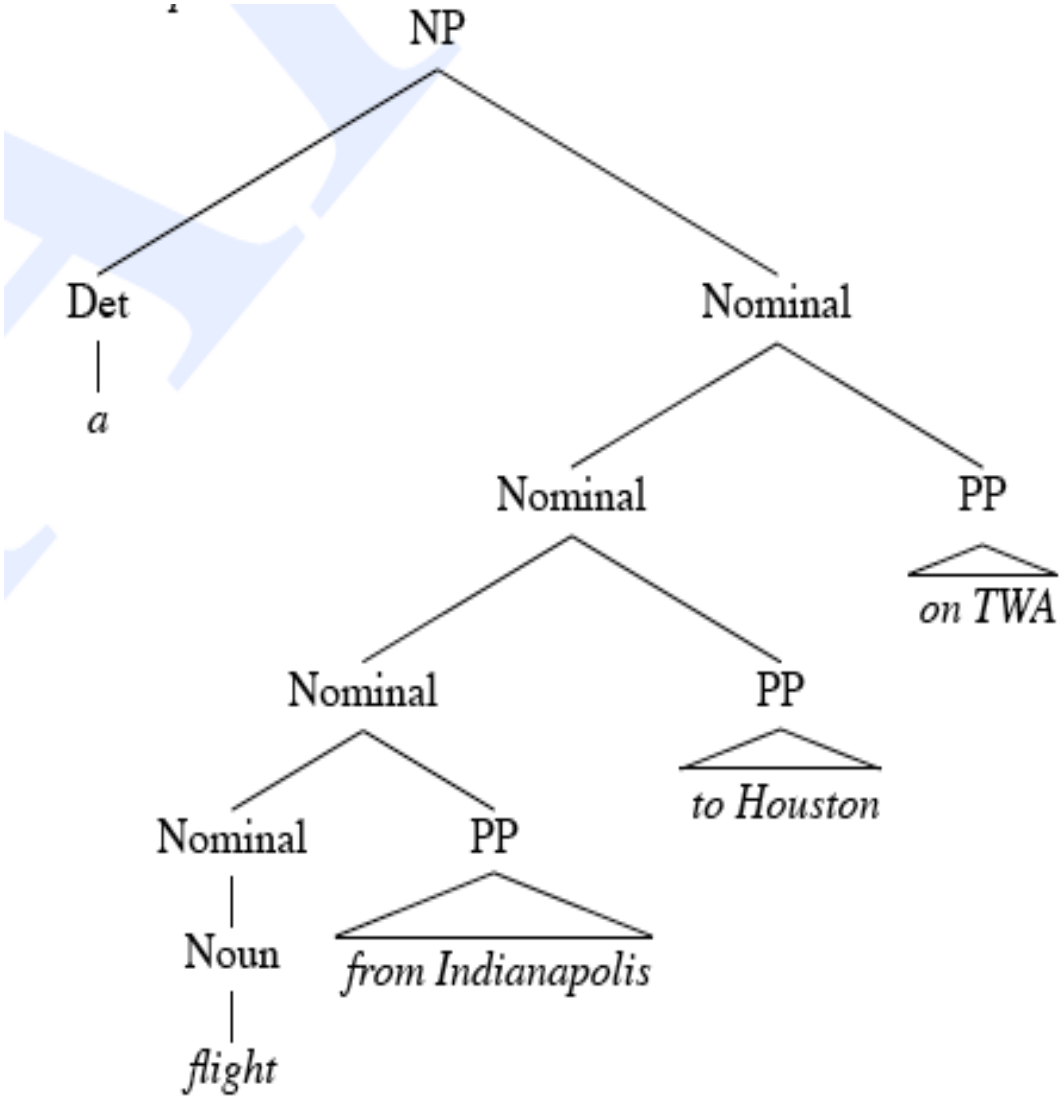
Shared Sub-problems



Shared Sub-problems



Shared Sub-problems



Structural ambiguity

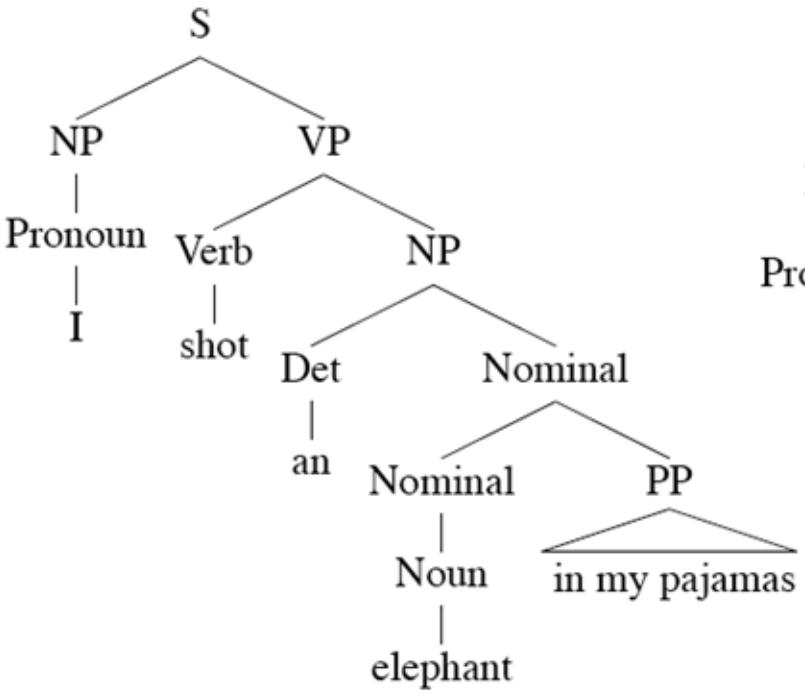
- One sentence can have several “legal parse tree”
- Structural (cf. PoS ambiguity)
  - attachment ambiguity
  - coordination ambiguity
- 15 words ⇒ ~1000000 parse trees

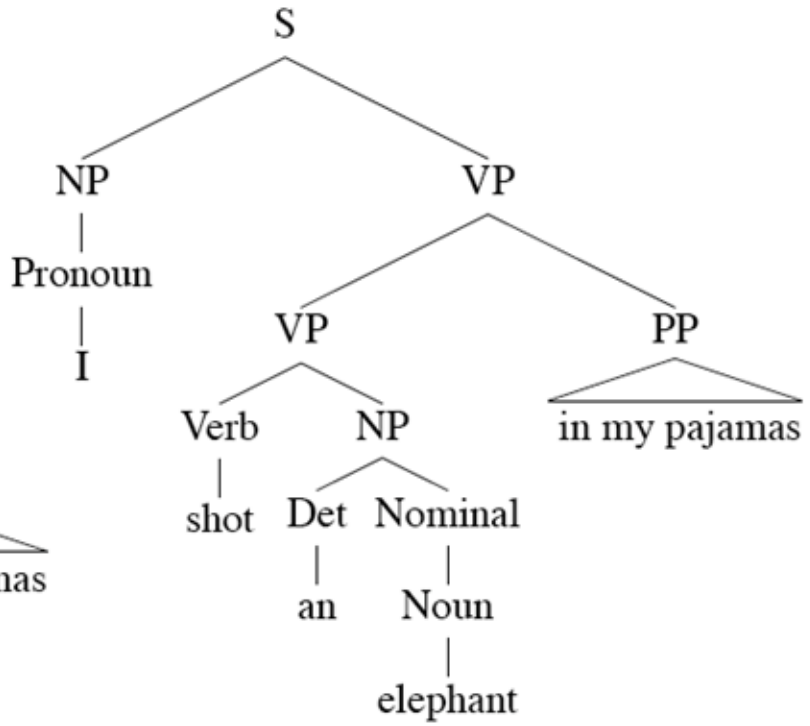
Attachment ambiguity (PP)

One morning I shot an elephant in my pajamas.

How he got into my pajamas I don't know.

Groucho Marx, *Animal Crackers*, 1930





Attachment ambiguity (PP)

Coordination ambiguity

... televisioni e radio nazionali ...

Si può mangiare e bere qualcosa

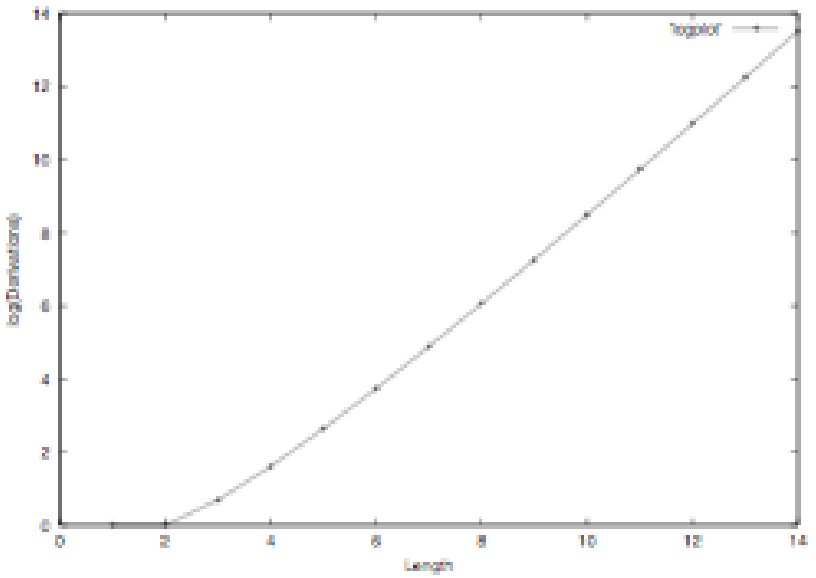
Si può mangiare e pagare qualcosa

Coordination ambiguity

Structural ambiguity

- CFG rules { S → S S , S → a }

#a	# parses
1	1
2	1
3	2
4	5
5	14
6	42
7	132
8	429
9	1430
10	4862
11	16796



Structural ambiguity

Catalan numbers (Church and Patil 1982)

- Serie potenza
  - N → natural | language | processing | course
  - N→NN

$$C_n = \frac{1}{n+1}\binom{2n}{n} = \frac{(2n)!}{(n+1)!\,n!} = \prod_{k=2}^{n}\frac{n+k}{k} \qquad \text{for } n \geq 0.$$

- Intuizione
  - ugual numero di parentesi aperte e chiuse
  - propriamente incapsulate, i.e. un'aperta precede una chiusa

Come gestire l'esplosione combinatoria?

Dynamic Programming

- **Richard Bellman, 1957** Principio di ottimalità: “An optimal policy has the property that whatever the initial state and initial decision are, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision”.
  - Sottostrutture ottimali
  - Sottoproblemi sovrapponibili
  - Memoizzazione
- Simile a dividi et impera

Come gestire l'esplosione combinatoria?

Dynamic Programming Parsing

- **CKY algorithm**: calcola tutti i possibili parse in tempo $O(n^3)$. Il difetto maggiore è che il caso peggiore e il caso medio coincidono (ma anche esige una forma normale per la grammatica).
- **Earley algorithm**: calcola tutti i possibili parse in tempo $O(n^3)$ per una CFG arbitraria, $O(n^2)$ per una CFG ambigua, e $O(n)$ per una **bounded-state CFG** (e.g. **S → aSa | bSb | aa | bb**). Il caso medio ha una performance migliore rispetto a CKY.
- **Deterministic parsing**: solo un parsing -> ambiguità limitata -> programming language parsing
  - top-down: e.g. LL parsing
  - bottom-up: LR or shift-reduce parsing

Outline

- Parser anatomy
- Top-down vs. bottom-up
- **CKY algorithm: Cocke-Kasami-Younger**
- Parsing probabilistico e TB grammars
- Parsing parziale

Parser B (CKY- Cocke Kasamy Younger)

(1) Grammar

Context-Free, ...

(2) Algorithm

I. Search strategy

top-down, bottom-up, left-to-right, ...

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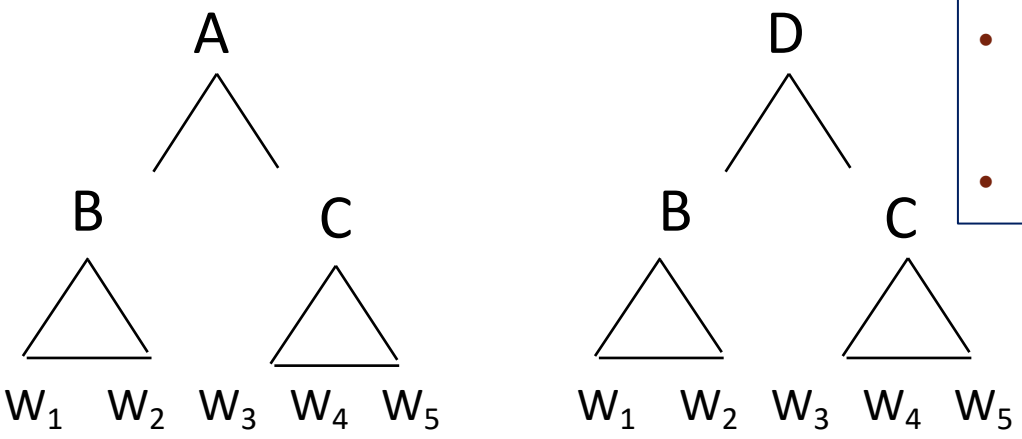
Probabilistic, rule-based, ...

CKY idea

$A \rightarrow B\ C$

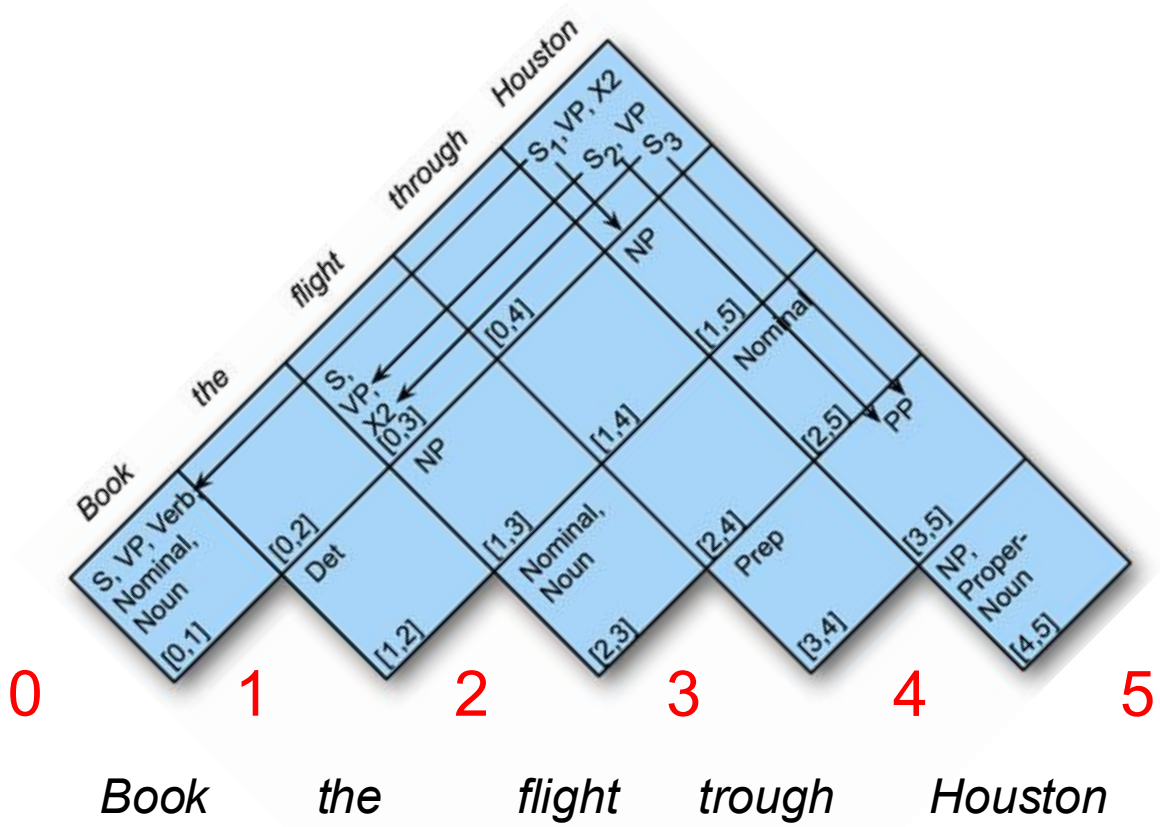
$D \rightarrow B\ C$



CF in Forma Normale di Chomsky

- Copy all conforming rules to the new grammar unchanged,
- Convert terminals within rules to dummy non-terminals,
- Convert unit-productions,
- Binarize all rules and add to new grammar.
- Epsilon free

CKY: idea della table



CKY $_0Book_1that_2flight_3$

A -> BC

- Se c'è un A allora c'è un B seguito da un C
- Se A va da **i to j**, c'è un k tale che **i<k<j**
- Cioè B va da **i to k** e C va da **k to j**
- Usiamo una **table** (array bidimensionale) per mettere **B in table[i,k]**, **C in table[k,j]**, A **in table[i,j]**
- Loop su k
- Se il parsing ha successo -> **S in [0,n]**

Grammatica L1

Grammar	Lexicon
$S \rightarrow NP\ VP$	$Det \rightarrow that \mid this \mid a$
$S \rightarrow Aux\ NP\ VP$	$Noun \rightarrow book \mid flight \mid meal \mid money$
$S \rightarrow VP$	$Verb \rightarrow book \mid include \mid prefer$
$NP \rightarrow Pronoun$	$Pronoun \rightarrow I \mid she \mid me$
$NP \rightarrow Proper\text{-}Noun$	$Proper\text{-}Noun \rightarrow Houston \mid NWA$
$NP \rightarrow Det\ Nominal$	$Aux \rightarrow does$
$Nominal \rightarrow Noun$	$Preposition \rightarrow from \mid to \mid on \mid near \mid through$
$Nominal \rightarrow Nominal\ Noun$	
$Nominal \rightarrow Nominal\ PP$	
$VP \rightarrow Verb$	
$VP \rightarrow Verb\ NP$	
$VP \rightarrow Verb\ NP\ PP$	
$VP \rightarrow Verb\ PP$	
$VP \rightarrow VP\ PP$	
$PP \rightarrow Preposition\ NP$	

Grammatica L1 e normal form

$S \rightarrow NP\ VP$	$S \rightarrow NP\ VP$
$S \rightarrow Aux\ NP\ VP$	$S \rightarrow X1\ VP$
	$X1 \rightarrow Aux\ NP$
$S \rightarrow VP$	$S \rightarrow book \mid include \mid prefer$
	$S \rightarrow Verb\ NP$
	$S \rightarrow X2\ PP$
	$S \rightarrow Verb\ PP$
	$S \rightarrow VP\ PP$
$NP \rightarrow Pronoun$	$NP \rightarrow I \mid she \mid me$
$NP \rightarrow Proper\text{-}Noun$	$NP \rightarrow TWA \mid Houston$
$NP \rightarrow Det\ Nominal$	$NP \rightarrow Det\ Nominal$
$Nominal \rightarrow Noun$	$Nominal \rightarrow book \mid flight \mid meal \mid money$
$Nominal \rightarrow Nominal\ Noun$	$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$	$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow Verb$	$VP \rightarrow book \mid include \mid prefer$
$VP \rightarrow Verb\ NP$	$VP \rightarrow Verb\ NP$
$VP \rightarrow Verb\ NP\ PP$	$VP \rightarrow X2\ PP$
	$X2 \rightarrow Verb\ NP$
$VP \rightarrow Verb\ PP$	$VP \rightarrow Verb\ PP$
$VP \rightarrow VP\ PP$	$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$	$PP \rightarrow Preposition\ NP$

CKY Algorithm

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
      for all {A | A → words[j] ∈ grammar}
              table[j − 1, j] ← table[j − 1, j] ∪ A
      for i ← from j − 2 down to 0 do
          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                      table[i,j] ← table[i,j] ∪ A
```

Figure 18.12 The CKY algorithm.

CKY Algorithm

```
function CKY-PARSE(words, grammar) returns table

  for j ← from 1 to LENGTH(words) do
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          for k ← i + 1 to j − 1 do
              for all {A | A → BC ∈ grammar and B ∈ table[i,k] and C ∈ table[k, j]}
                    table[i,j] ← table[i,j] ∪ A
```

Figure 18.12 The CKY algorithm.

Looping over the columns

Filling the bottom cell

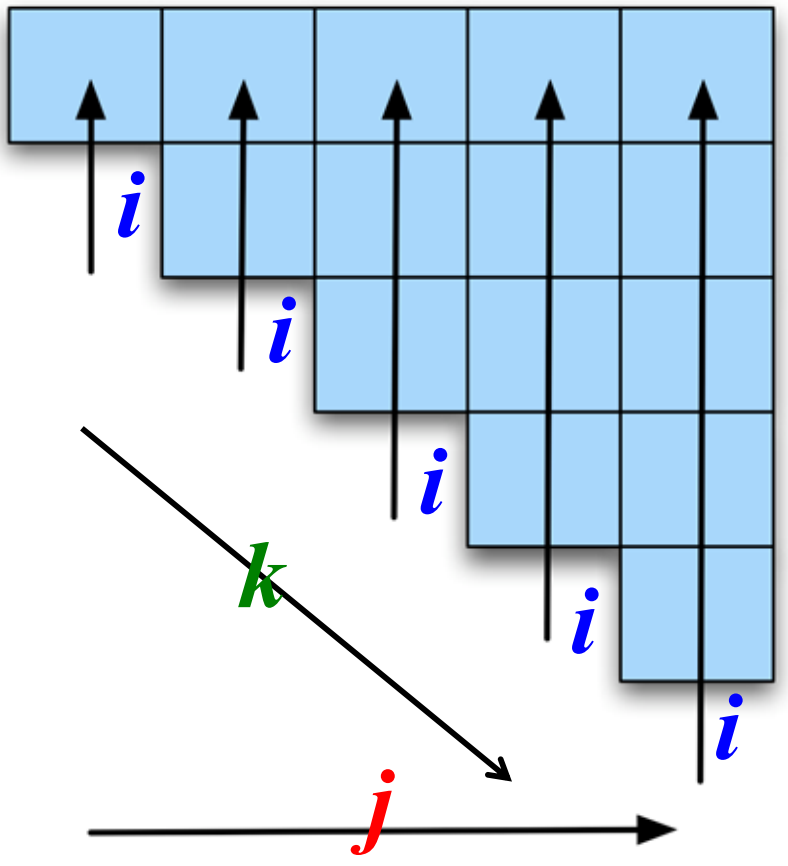
Filling row i in column j

Looping over the possible split locations between i and j.

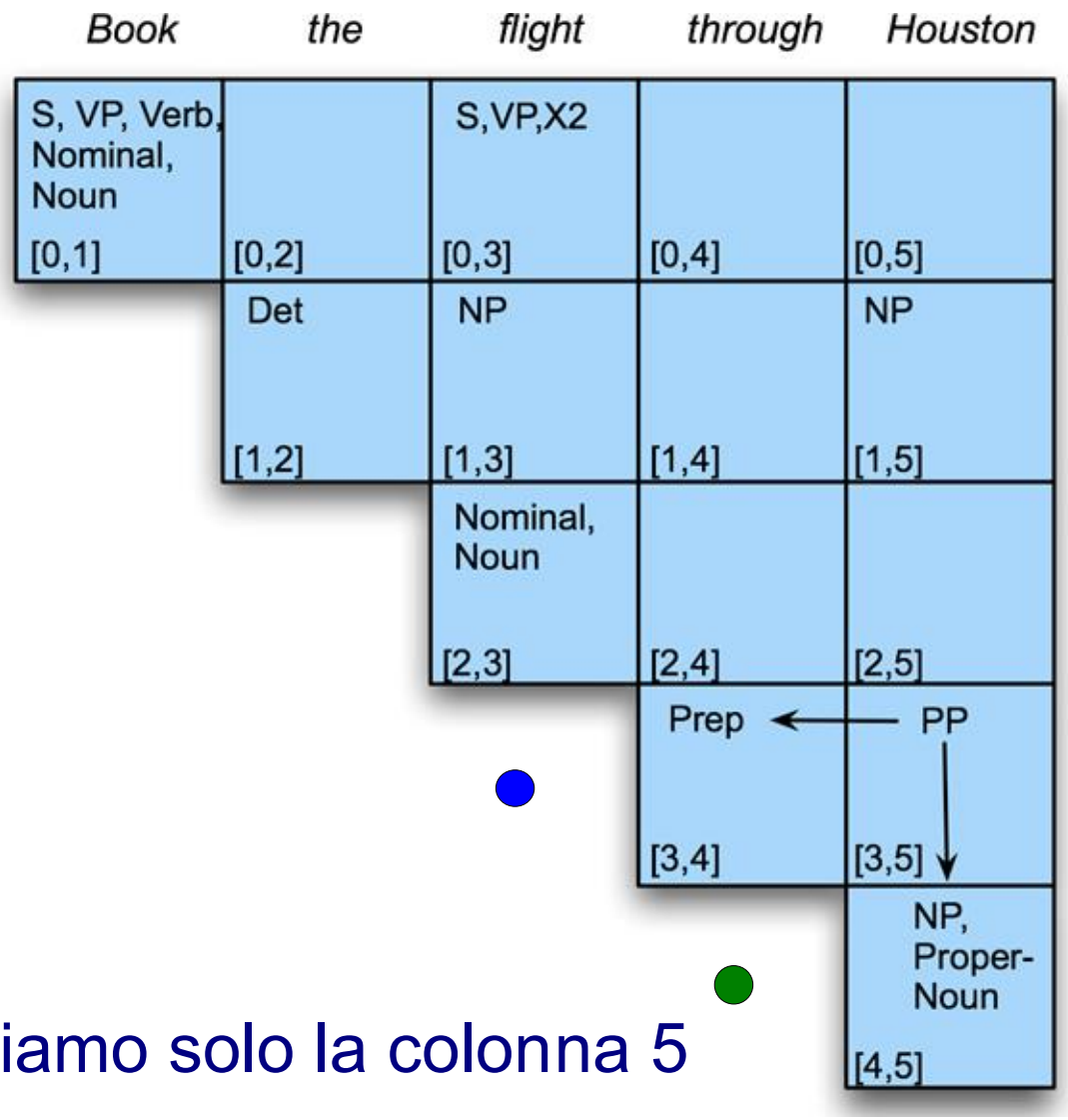
Check the grammar for rules that link the constituents in [i,k] with those in [k,j]. For each rule found store the LHS of the rule in cell [i,j].

$[\ j : 1...n\ [\ i : j\text{-}2...0\ [\ k : i{+}1...j{-}1\ ]\ ]\ ]$

	Book	the	flight	through	Houston
0	S, VP, Verb Nominal, Noun [0,1]	[0,2]	S,VP,X2 [0,3]	[0,4]	S,VP,X2 [0,5]
1	1	Det [1,2]	NP [1,3]	[1,4]	NP [1,5]
2		2	Nominal, Noun [2,3]	[2,4]	Nominal [2,5]
3			3	Prep [3,4]	PP [3,5]
				4	NP, Proper-Noun [4,5]
	1	2	3	4	5



$[\ j : 1...n\ [\ i : j\text{-}2...0\ [\ k : i{+}1...j{-}1\ ]\ ]\ ]$



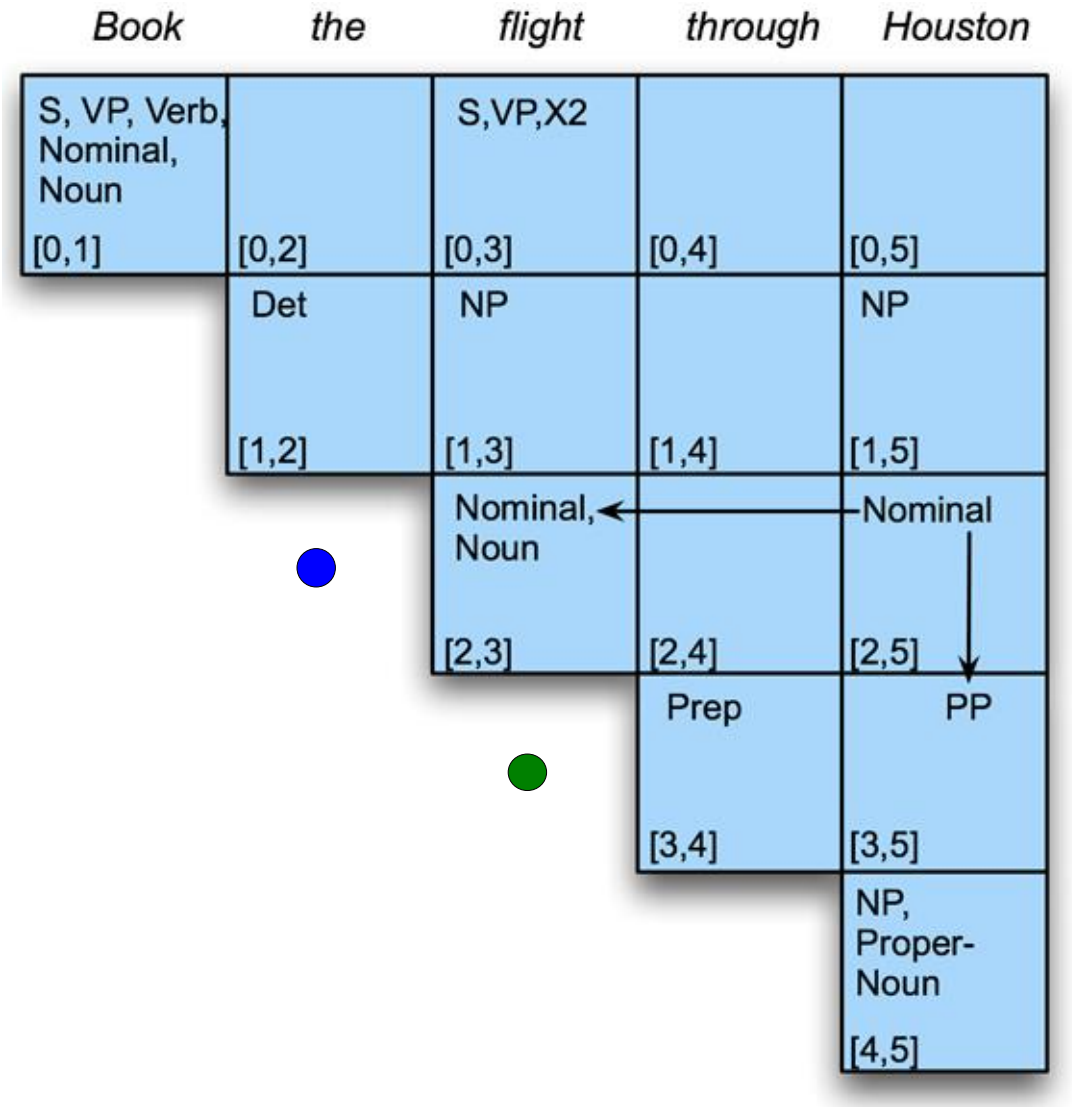
j=5

i=3

k=4

Consideriamo solo la colonna 5

$[\, j : 1...n \; [\, i : j\text{-}2...0 \; [\, k : i{+}1...j\text{-}1 \,]\,]\,]$

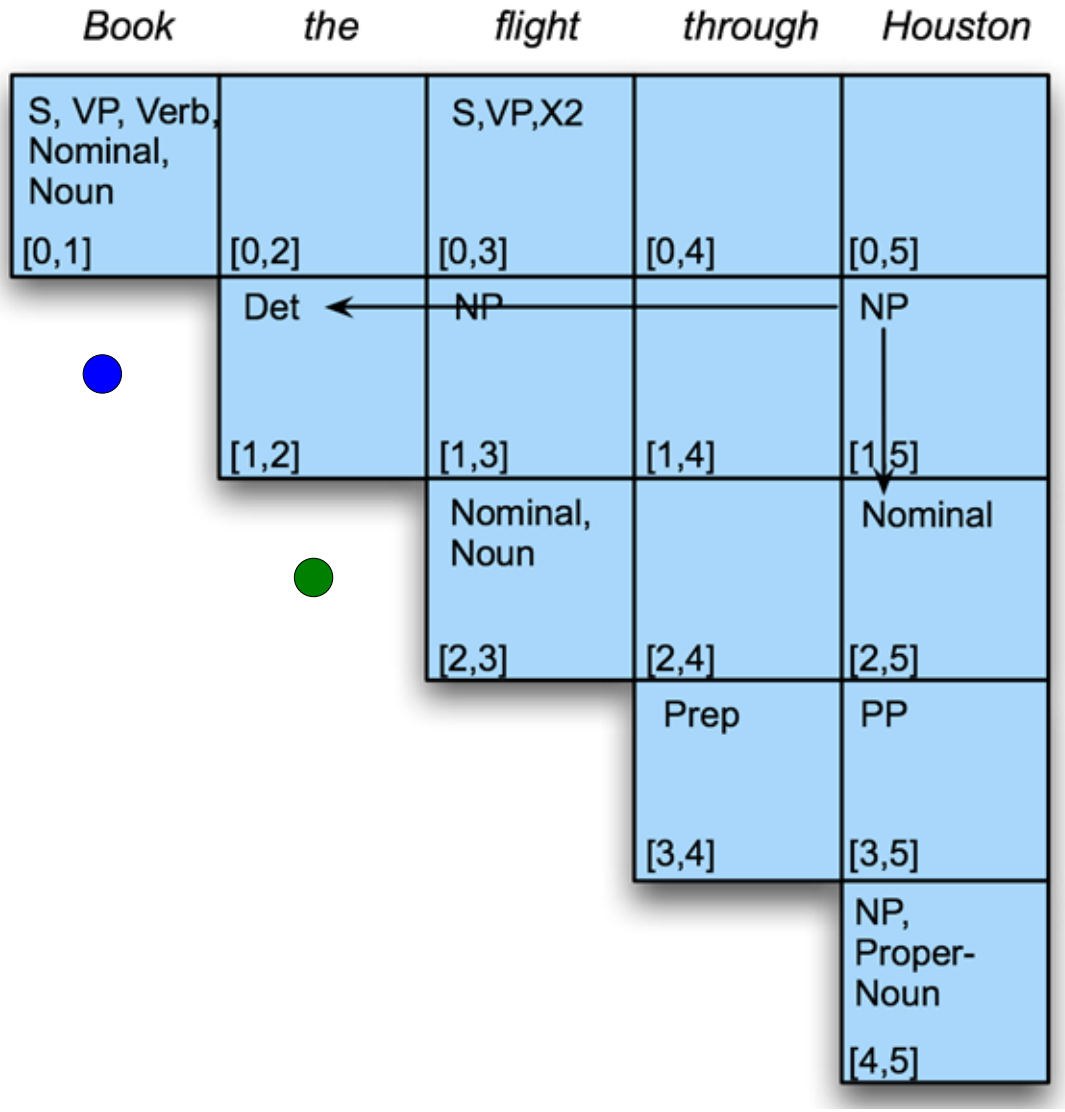


j=5

i=2

k=3

$[\ j : 1...n\ [\ i : j\text{-}2...0\ [\ k : i{+}1...j\text{-}1\ ]\ ]\ ]$

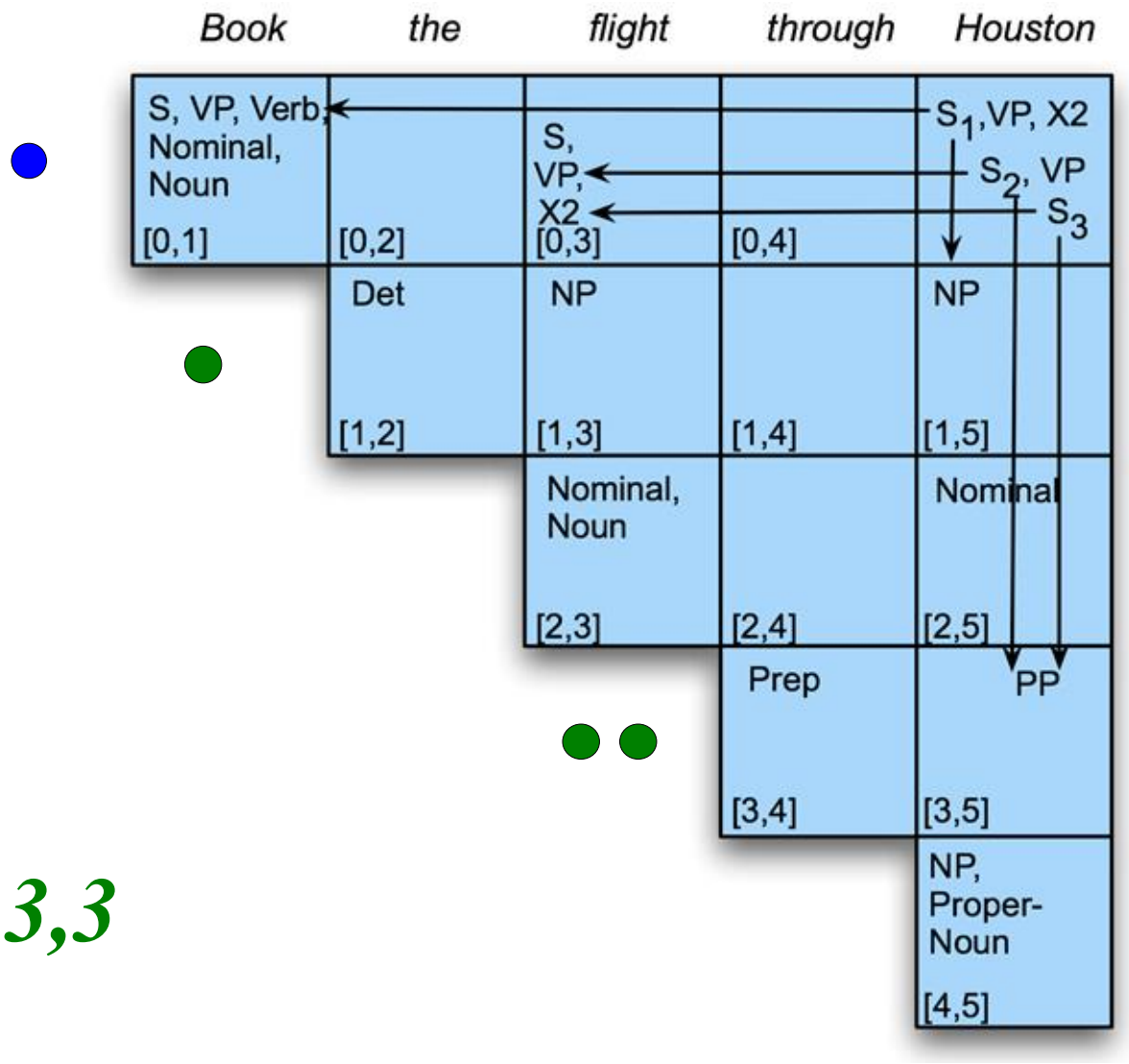


j=5

i=1

k=2

$[\ j : 1...n\ [\ i : j\text{-}2...0\ [\ k : i{+}1...j{-}1\ ]\ ]\ ]$

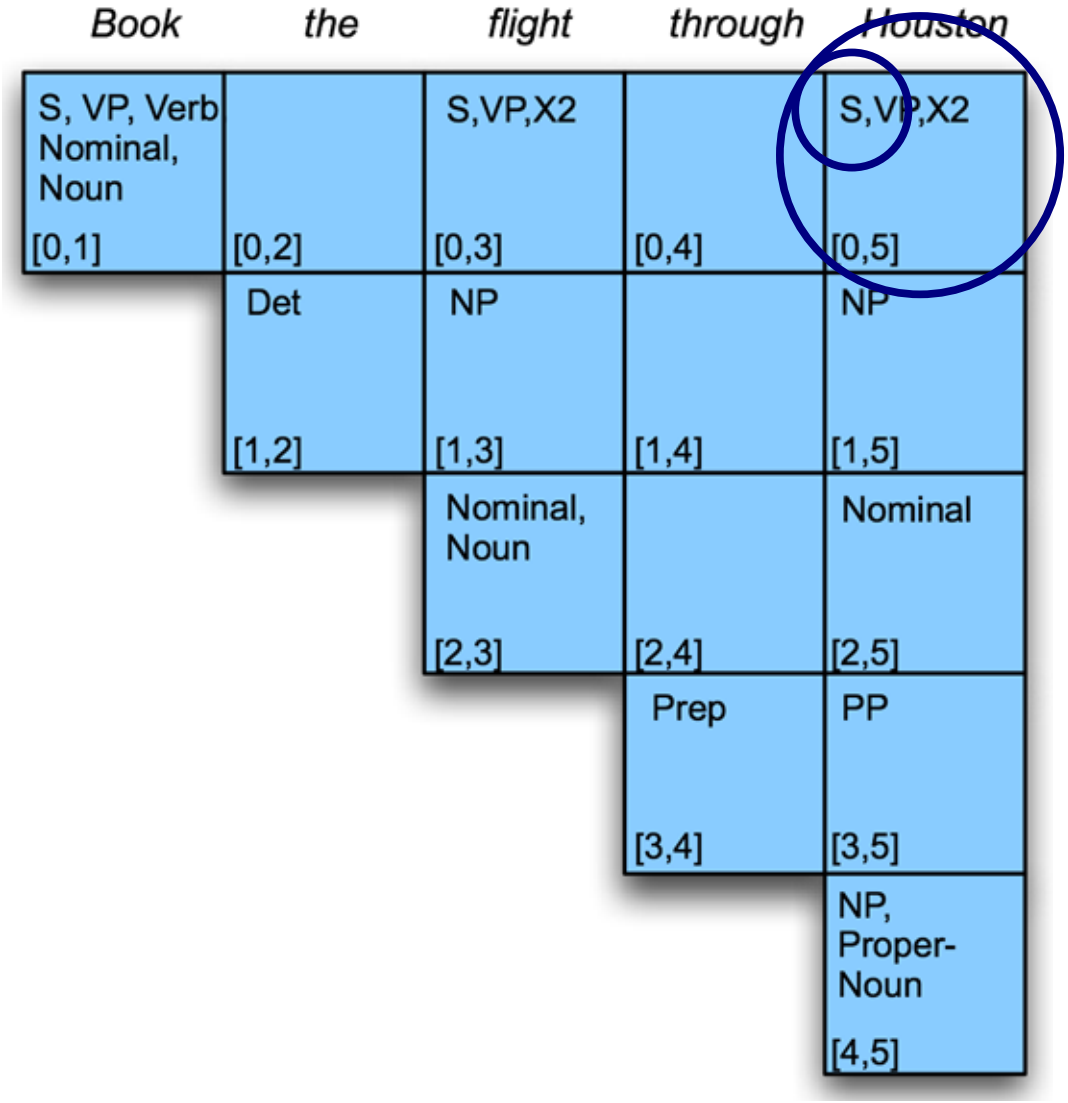


j=5

i=0

k=1,3,3

Successo!


Book
the
flight
through
Houston
S, VP, Verb Nominal, Noun
[0,1]
[0,2]
S,VP,X2
[0,3]
[0,4]
S,VP,X2
[0,5]
Det
[1,2]
NP
[1,3]
[1,4]
NP
[1,5]
Nominal, Noun
[2,3]
[2,4]
Nominal
[2,5]
Prep
[3,4]
PP
[3,5]
NP, Proper-Noun
[4,5]

Parsing vs. recognition -> backtrace

Attenzione!!!

Complessità

- $O(n^3)$
- $O(n^5)$ per la versione probabilistica (vedi dopo)
- Troppo lento per il real-time (motori di ricerca)
- Allora:
  - parziale
  - pruning probabilistico
  - dipendenze (prossima lezione)

Bisogna provare ...

S -> NP VP
VP -> VP ADV
VP -> V NP

NP -> Paolo
NP -> Francesca
V -> ama
ADV -> dolcemente

$_0Paolo_1ama_2Francesca_3dolcemente_4$

Outline

- Parser anatomy
- Top-down vs. bottom-up
- CKY algorithm: Cocke-Kasami-Younger
- **Parsing probabilistico e TB grammars**
- Parsing parziale

Parser C (CKY probabilistico)

(1) Grammar

Probabilistic Context-Free, ...

(2) Algorithm

I. Search strategy

top-down, bottom-up, left-to-right, ...

II. Memory organization

back-tracking, dynamic programming (+beam?), ...

(3) Oracle

Probabilistic, rule-based, ...

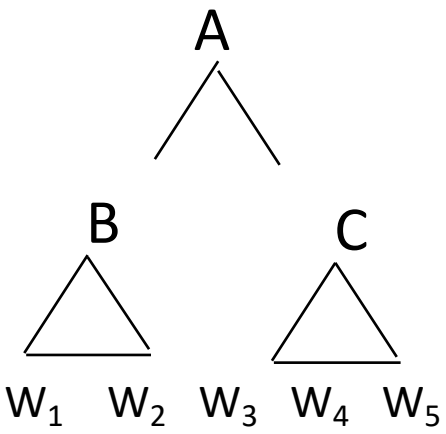
CKY probabiistico

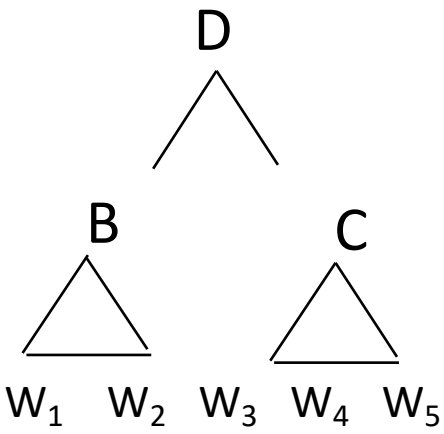
$A \rightarrow BC \quad [p_A]$

$D \rightarrow BC \quad [p_D]$

$$P(1,4,A) = p_A * P(1,2,B) * P(3,4,C)$$

$$P(1,4,D) = p_D * P(1,2,B) * P(3,4,C)$$





Probabilistic CFG

G=(Σ,V,S,P)

$A \rightarrow \beta$ [p] $p \in (0,1)$

$$\sum_{\beta} P(A \rightarrow \beta) = 1$$

PCFG

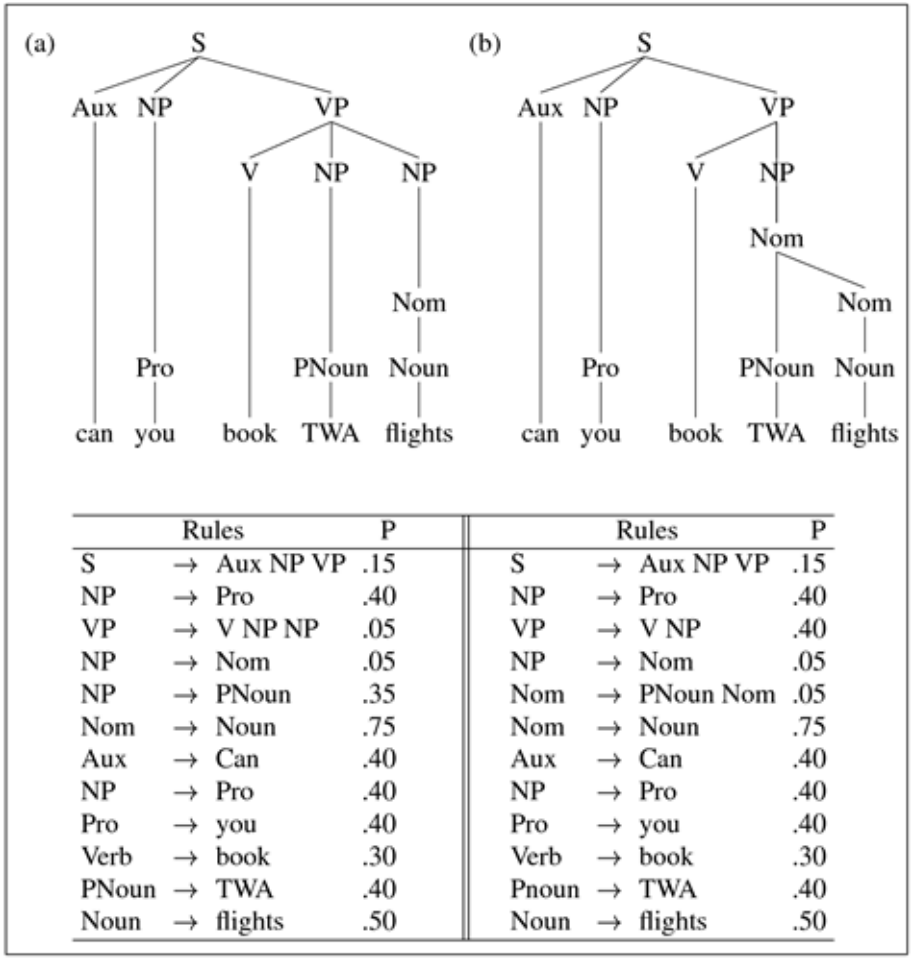
$S \rightarrow NP\ VP$	[.80]	$Det \rightarrow that$ [.05] | the [.80] | a [.15]	
$S \rightarrow Aux\ NP\ VP$	[.15]	$Noun \rightarrow book$	[.10]
$S \rightarrow VP$	[.05]	$Noun \rightarrow flights$	[.50]
$NP \rightarrow Det\ Nom$	[.20]	$Noun \rightarrow meal$	[.40]
$NP \rightarrow Proper\text{-}Noun$	[.35]	$Verb \rightarrow book$	[.30]
$NP \rightarrow Nom$	[.05]	$Verb \rightarrow include$	[.30]
$NP \rightarrow Pronoun$	[.40]	$Verb \rightarrow want$	[.40]
$Nom \rightarrow Noun$	[.75]	$Aux \rightarrow can$	[.40]
$Nom \rightarrow Noun\ Nom$	[.20]	$Aux \rightarrow does$	[.30]
$Nom \rightarrow Proper\text{-}Noun\ Nom$	[.05]	$Aux \rightarrow do$	[.30]
$VP \rightarrow Verb$	[.55]	$Proper\text{-}Noun \rightarrow TWA$	[.40]
$VP \rightarrow Verb\ NP$	[.40]	$Proper\text{-}Noun \rightarrow Denver$	[.40]
$VP \rightarrow Verb\ NP\ NP$	[.05]	$Pronoun \rightarrow you$ [.40] | I [.60]	

Modello probabilistico: ipotesi

La probabilità di un albero è il prodotto delle regole usate nella sua derivazione

$$P(T,S) = \prod_{node \in T} P(rule(n))$$

PCFG



Rules			P
S	→	Aux NP VP	.15
NP	→	Pro	.40
VP	→	V NP NP	.05
NP	→	Nom	.05
NP	→	PNoun	.35
Nom	→	Noun	.75
Aux	→	Can	.40
NP	→	Pro	.40
Pro	→	you	.40
Verb	→	book	.30
PNoun	→	TWA	.40
Noun	→	flights	.50

Rules			P
S	→	Aux NP VP	.15
NP	→	Pro	.40
VP	→	V NP	.40
NP	→	Nom	.05
Nom	→	PNoun Nom	.05
Nom	→	Noun	.75
Aux	→	Can	.40
NP	→	Pro	.40
Pro	→	you	.40
Verb	→	book	.30
Pnoun	→	TWA	.40
Noun	→	flights	.50

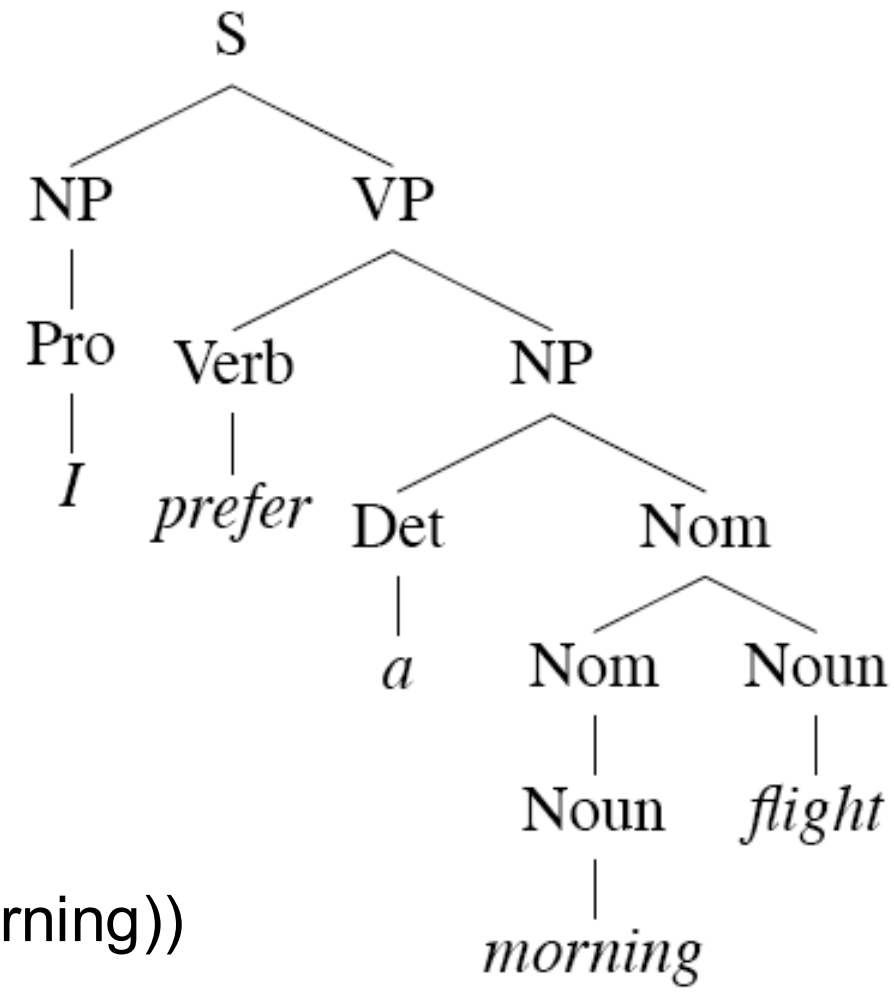
$P(T_a) = .15 * .4 * .05 * .05 * .35 * .75 * .4 * .4 * .4 * .3 * .4 * .5 = 1.5 \times 10^{-6}$

$P(T_b) = .15 * .4 * .4 * .05 * .05 * .75 * .4 * .4 * .4 * .3 * .4 * .5 = 1.7 \times 10^{-6}$

Da dove prendo le probabilità?

- Treebanks -> banche di alberi
  - Costituenti -> Penn TB
  - Dipendenze -> TUT TB -> Universal dependency TB
- Si creano con sistemi semiautomatici:
  - prima un parser
  - poi correggo a mano
- Guidelines di annotazione
- Corpus linguistics

S-espressioni e alberi




```
(S (NP (Pro I))
    (VP (Verb prefer)
        (NP (Det a)
          (Nom (Nom (Noun morning))
                (Noun flight)))))
```

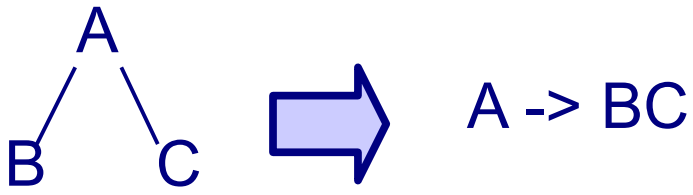
Penn TB

Wall Street Journal section of the Penn TreeBank. 1 M words from the 1987-1989 Wall Street Journal.

```
( (S (`` ``)
    (S-TPC-2
      (NP-SBJ-1 (PRP We) )
      (VP (MD would)
        (VP (VB have)
          (S
            (NP-SBJ (-NONE- *-1) )
            (VP (TO to)
              (VP (VB wait)
                (SBAR-TMP (IN until)
                  (S
                    (NP-SBJ (PRP we) )
                    (VP (VBP have)
                      (VP (VBN collected)
                        (PP-CLR (IN on)
                          (NP (DT those)(NNS assets))))))))))))
    (, ,) (`` ``)
    (NP-SBJ (PRP he) )
    (VP (VBD said)
      (S (-NONE- *T*-2) ))
    (. .) ))
```

TB grammars

- I TB implicitamente definiscono delle grammatiche CF
  - induzione (?)
- Regole **locali**:



- WSJ -> 12k regole (!!)
- A volte regole *flat*, gli annotatori evitano la ricorsione:

```
VP → VBD PP
VP → VBD PP PP
VP → VBD PP PP PP
VP → VBD PP PP PP PP
```

Modello probabilistico: training

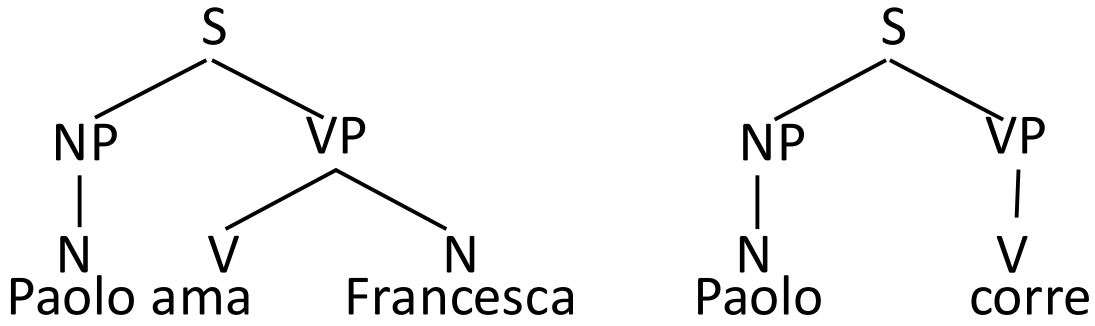
- Abbiamo bisogno di

$$P(\alpha \rightarrow \beta|\alpha)$$

- Quindi, per ogni albero nel TB, calcoliamo:

$$P(\alpha \rightarrow \beta|\alpha) = \frac{\text{Count}(\alpha \rightarrow \beta)}{\sum_{\gamma}\text{Count}(\alpha \rightarrow \gamma)} = \frac{\text{Count}(\alpha \rightarrow \beta)}{\text{Count}(\alpha)}$$

Treebank Grammars (PCFG)



P(A→β)=Count(A→β)/Count(A)

P(S→NP VP) =2/2=1 P(NP→N) =2/2=1

P(VP→V N) =1/2=.5 P(VP→V) =1/2=.5

P(N→Paolo) =2/3=.66 P(N→Francesca) =1/3=.33

P(V→corre) =1/2=.5 P(V→ama) =1/2=.5

Parser C: CKY probabilistico

function PROBABILISTIC-CKY(*words*,*grammar*) **returns** most probable parse
and its probability

for $j \leftarrow$ **from** 1 **to** LENGTH(*words*) **do**
  for all $\{ A \mid A \rightarrow words[j] \in grammar \}$
    $table[j-1, j, A] \leftarrow P(A \rightarrow words[j])$
  for $i \leftarrow$ **from** $j-2$ **downto** 0 **do**
    for $k \leftarrow i+1$ **to** $j-1$ **do**
      for all $\{ A \mid A \rightarrow BC \in grammar,$
        and $table[i,k,B] > 0$ **and** $table[k,j,C] > 0 \}$
        if $(table[i,j,A] < P(A \rightarrow BC) \times table[i,k,B] \times table[k,j,C])$ **then**
          $table[i,j,A] \leftarrow P(A \rightarrow BC) \times table[i,k,B] \times table[k,j,C]$
          $back[i,j,A] \leftarrow \{k, B, C\}$
  return BUILD_TREE(*back*[1, LENGTH(*words*), *S*]), *table*[1, LENGTH(*words*), *S*]

Parser D: CKY Neurale (2019)

(1) Grammar

NN, …

(2) Algorithm

I. Search strategy

top-down, bottom-up, left-to-right, …

II. Memory organization

back-tracking, dynamic programming (+beam?), …

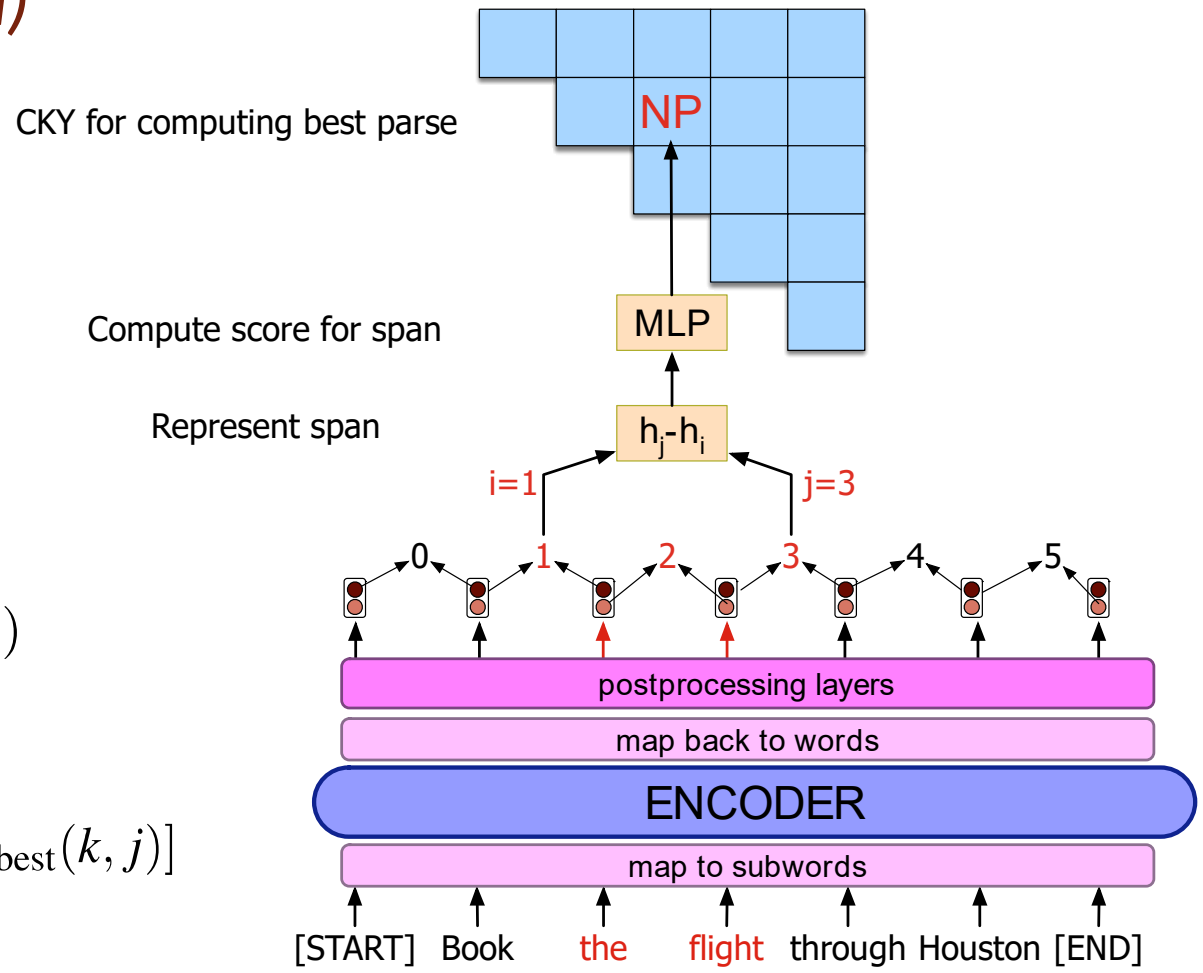
(3) Oracle

Probabilistic, rule-based, …

Parser D: CKY Neurale (2019)

- Span score: *s(l,j,l)*

$$s(T) = \sum_{(i,j,l) \in T} s(i,j,l)$$

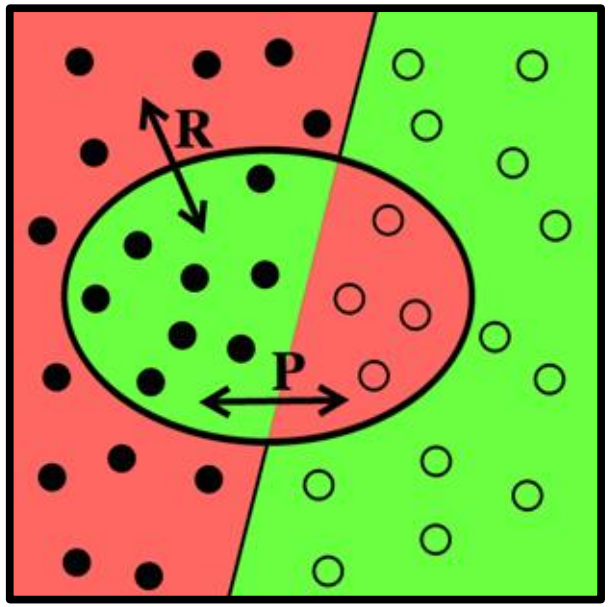


$$s_{\text{best}}(i, i+1) = \max_l s(i, i+1, l)$$

$$s_{\text{best}}(i,j) = \max_l s(i,j,l) + \max_k [s_{\text{best}}(i,k) + s_{\text{best}}(k,j)]$$

Valutazione

- Accuracy
- Precision vs. Recall
- F-score
- Unlabelled vs. Labelled



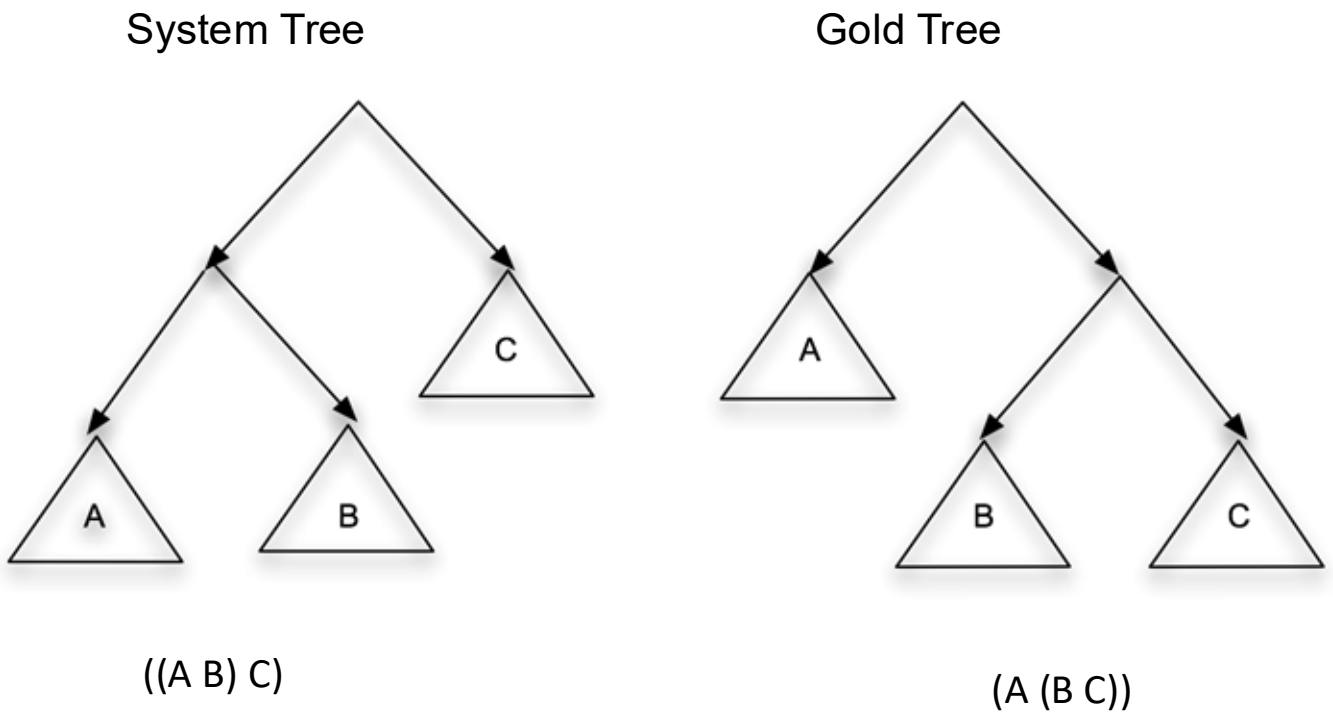
$$P = \frac{T\bullet}{T\bullet + F\bullet} \qquad R = \frac{T\bullet}{T\bullet + F\circ}$$

Valutazione per le PCFG

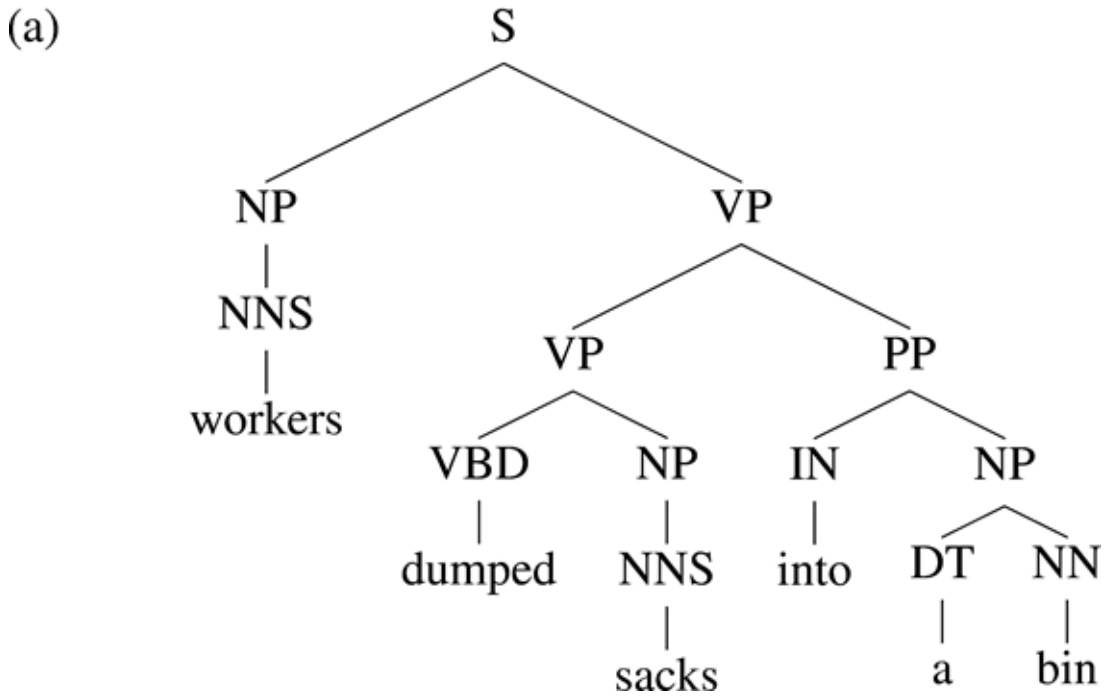
- Precision
  - Quale percentuale di subtree del system tree sono anche nel gold tree?
  - *Quanto di quello prodotto è giusto?*
- Recall
  - Quale percentuale di subtree del golden tree sono anche nel system tree?
  - *Quanto di quello che avremmo dovuto produrre abbiamo realmente prodotto?*

Parseval

- Crossing brackets

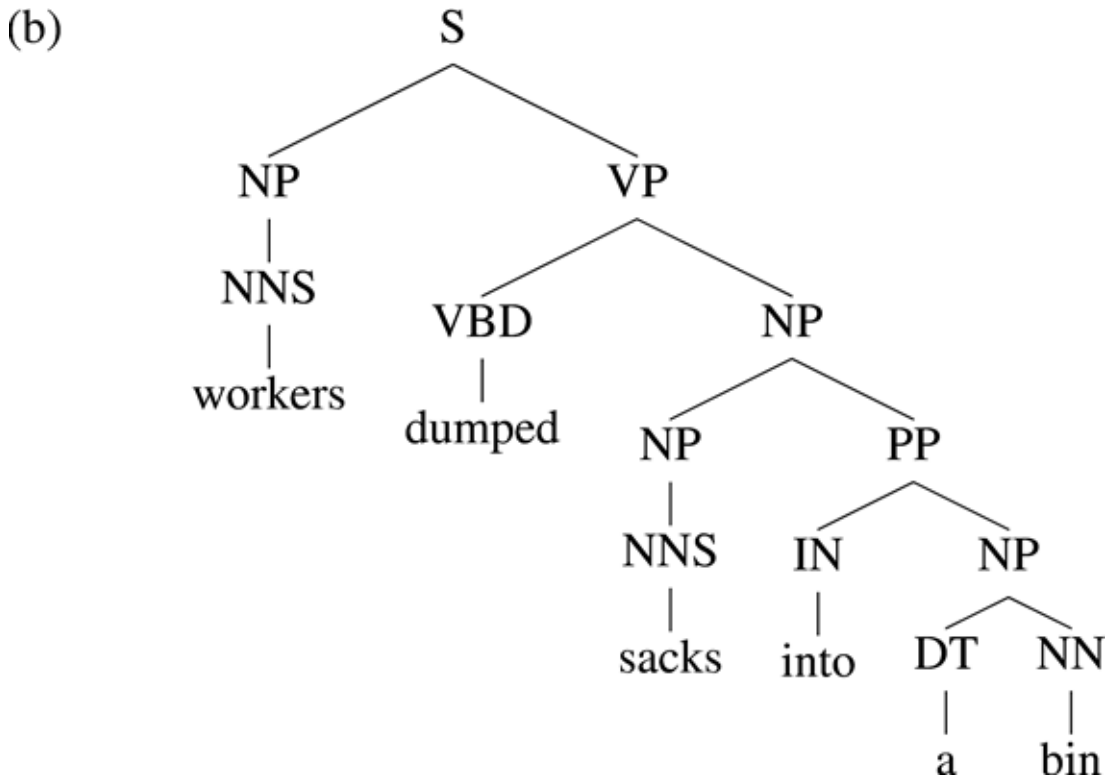


Problem with PCFG



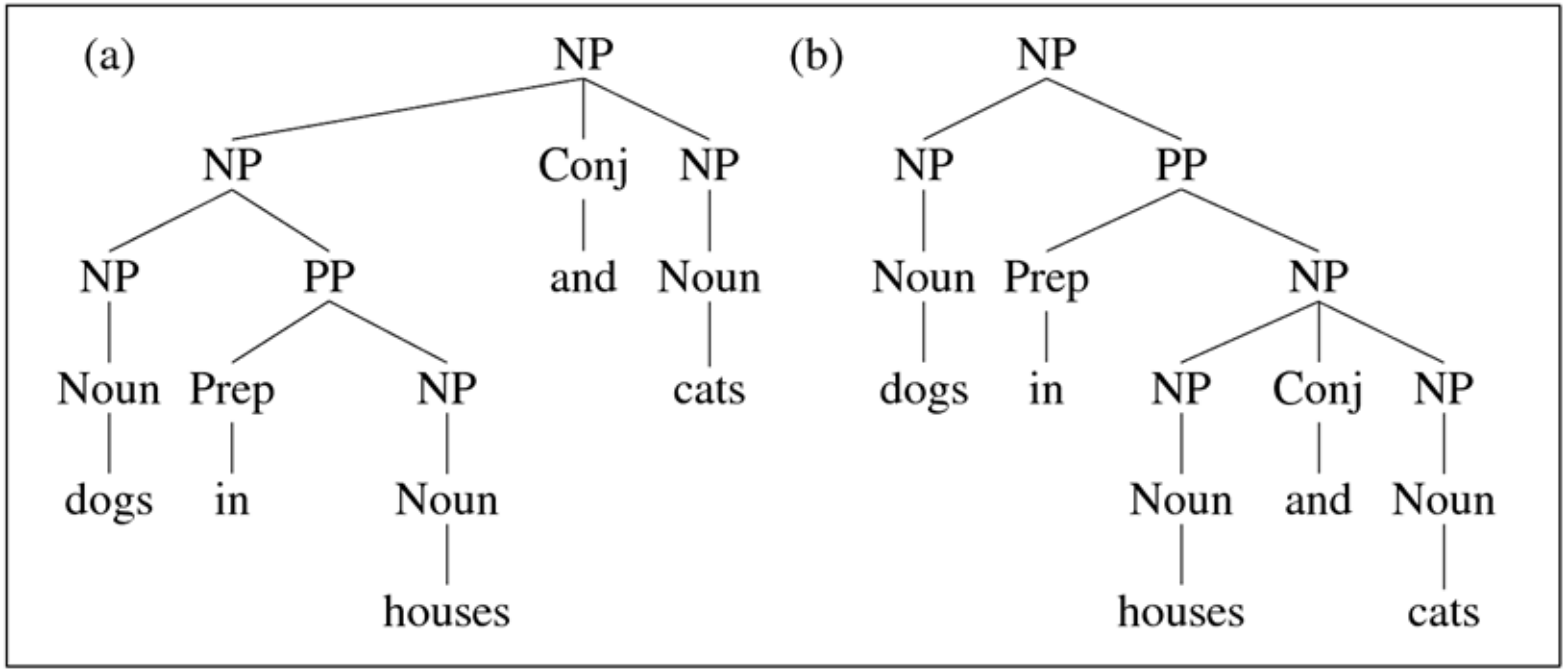
Ipotesi di indipendenza -> *non ci sono preferenze lessicali*

Problem with PCFG



Ipotesi di indipendenza -> *non ci sono preferenze lessicali*

Problem with PCFG



Ipotesi di indipendenza -> *non ci sono preferenze lessicali*

Lexicalized PCFG

Ogni regola CF è potenziata con informazione circa le **heads** dei costituenti coinvolti

$A \rightarrow BC$ $\qquad$ $A(head_A) \rightarrow B(head_B)\ C(head_C)$

Punto di unione tra formalismi sintattici a costituenti e a dipendenza

Lexicalized PCFG

VP → VBD NP PP

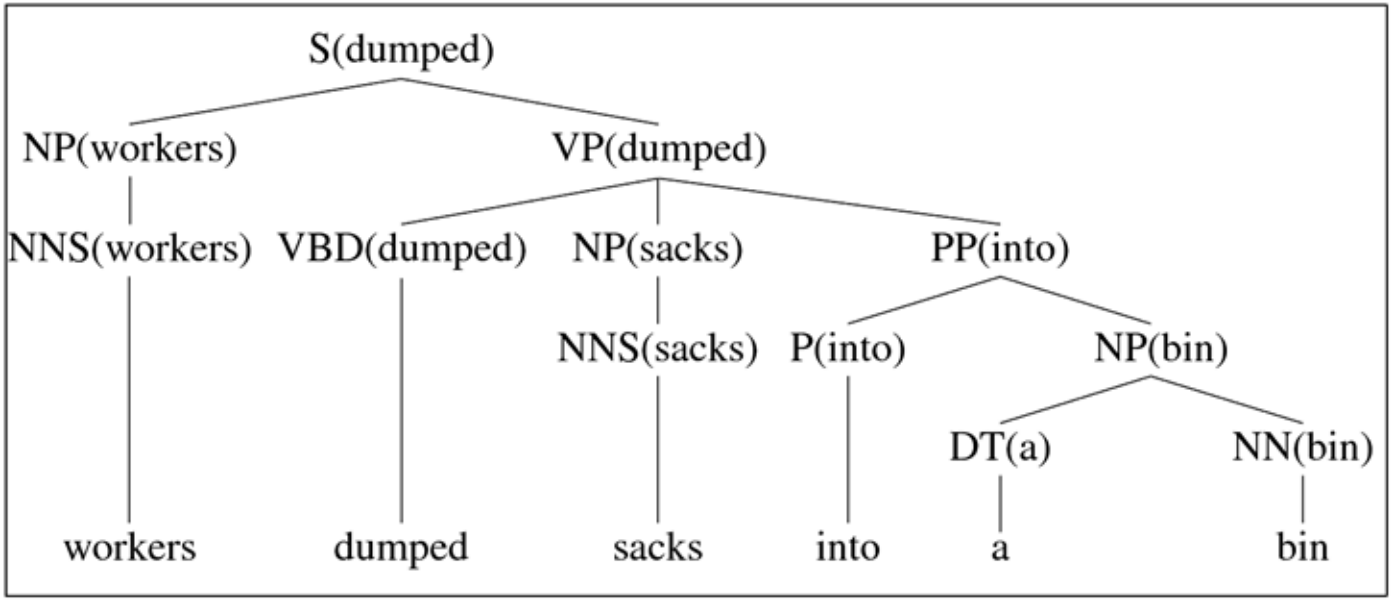
VP(dumped) → VBD(dumped) NP(sacks) PP(into)
[3×10^{-10}]

VP(dumped) → VBD(dumped) NP(cats) PP(into)
[8×10^{-11}]

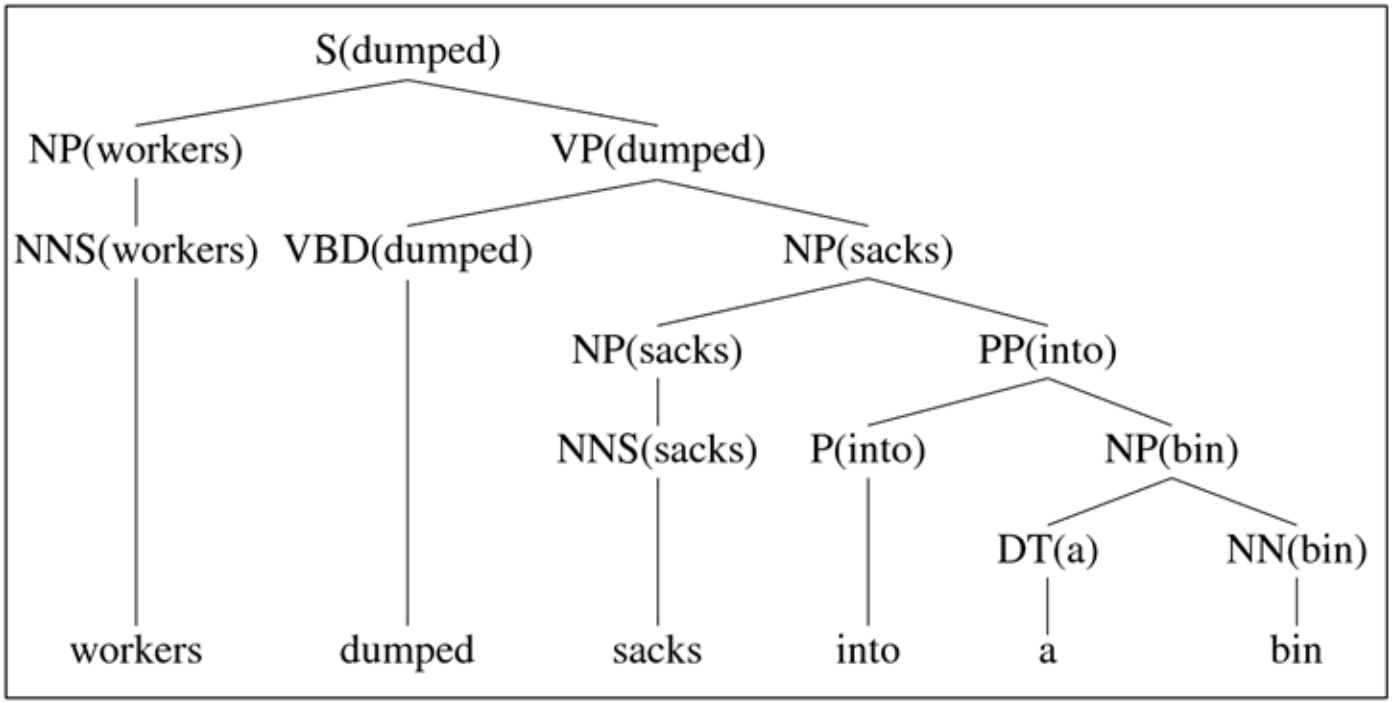
VP(dumped) → VBD(dumped) NP(hats) PP(into)
[4×10^{-10}]

VP(dumped)→VBD(dumped) NP(sacks) PP(above)
[1×10^{-12}]

Lexicalized PCFG



Lexicalized PCFG



Outline

- Parser anatomy
- Top-down vs. bottom-up
- CKY algorithm: Cocke-Kasami-Younger
- Parsing probabilistico e TB grammars
- **Parsing parziale**

Parser E (Chunk parsing a regole)

(1) Grammar

Regular-Grammars (cascade), ...

(2) Algorithm

I. Search strategy

top-down, bottom-up, left-to-right, ...

II. Memory organization

back-tracking, dynamic programming, ...

(3) Oracle

Probabilistic, rule-based, ...

Parsing parziale -> chunk parsing

- Base syntactic units-> CHUNK
  - **chuncks=basic non-recursive phrases**
- Elimino la ricorsione!

[$_{NP}$ The morning flight] [$_{PP}$ from] [$_{NP}$ Denver] [$_{VP}$ has arrived.]

[$_{NP}$ a flight] [$_{PP}$ from] [$_{NP}$ Indianapolis][$_{PP}$ to][$_{NP}$ Houston][$_{PP}$ on][$_{NP}$ TWA]

[$_{NP}$ The morning flight] from [$_{NP}$ Denver] has arrived.

Chunk parsing a regole

- Escludo la ricorsione: i sintagmi possono essere ricorsivi **i chunk NO**
- Cascata di regole gerarchiche

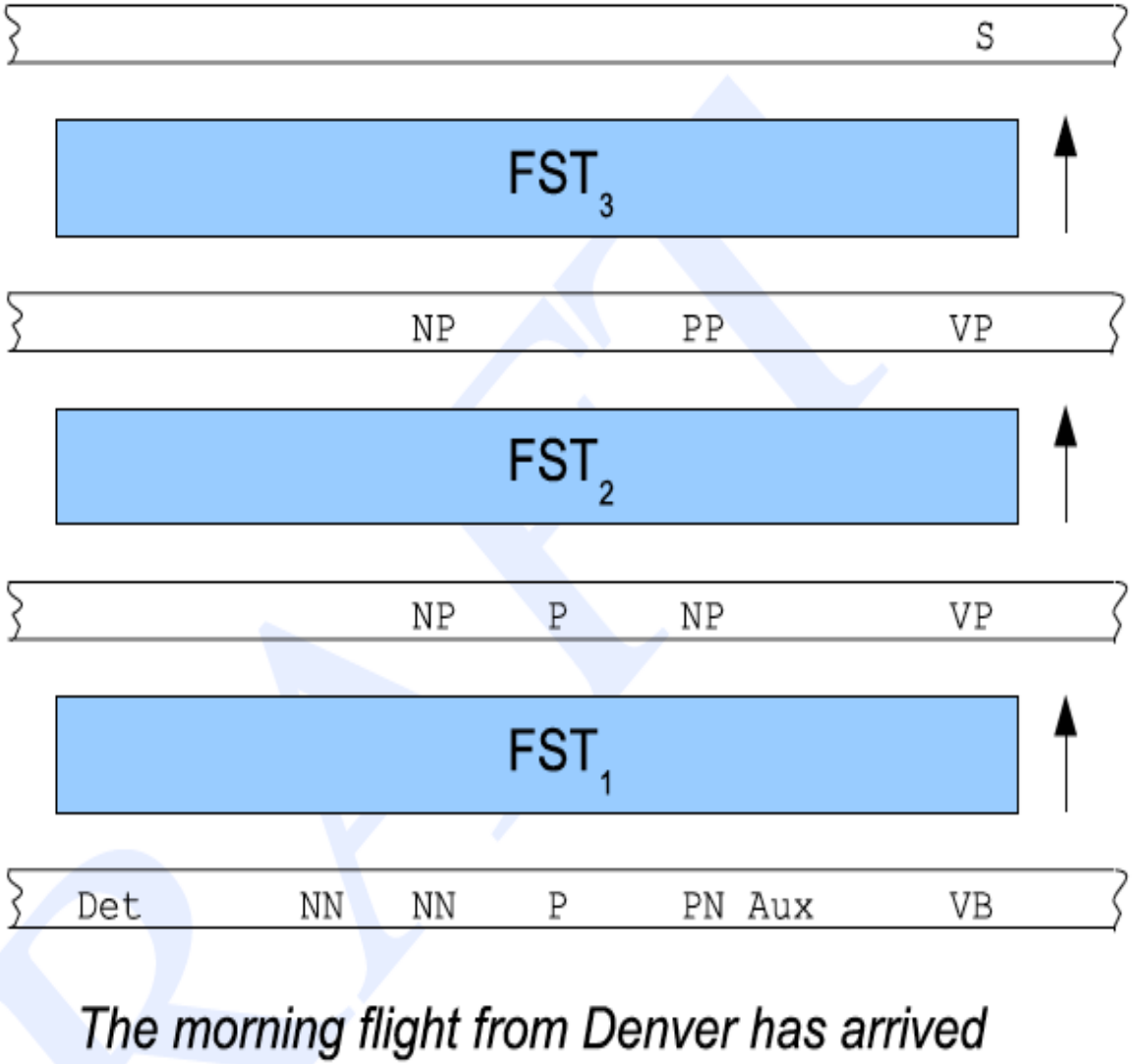
$$NP \rightarrow (Det)\ Noun^*\ Noun$$

$$NP \rightarrow Proper\text{-}Noun$$

$$VP \rightarrow Verb$$

$$VP \rightarrow Aux\ Verb$$

Cascata di trasducers: approssimo una CFG



Parser E (Chunk parsing probabilistico)

(1) Grammar

Regular-Grammars (cascade), ...

(2) Algorithm

I. Search strategy

top-down, bottom-up, left-to-right, ...

II. Memory organization

back-tracking, greedy (depth first), ...

(3) Oracle

Probabilistic, rule-based, ...

BIO tagging

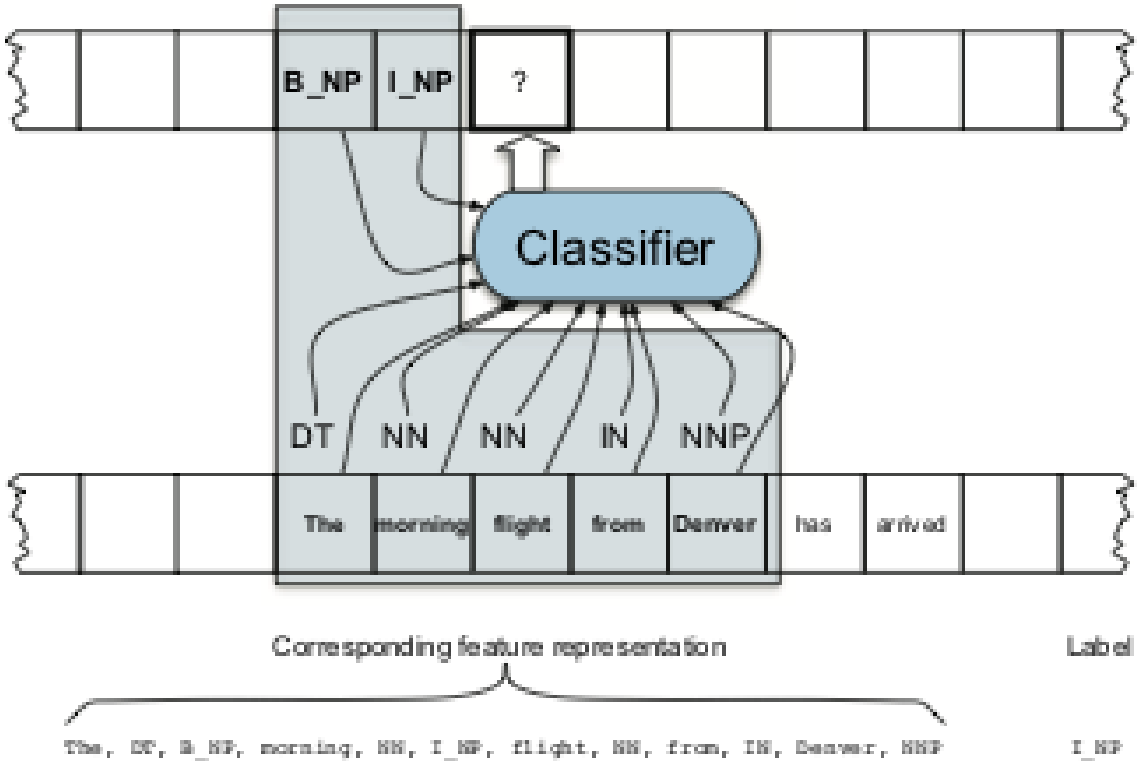
- Chunck parsing -> tagging
- *n* chuncks -> *2n+1* tags
- **B**egin , **I**nside, **O**utside

The morning flight from Denver has arrived
B_NP I_NP I_NP B_PP B_NP B_VP I_VP

The morning flight from Denver has arrived.
B_NP I_NP I_NP O B_NP O O

BIO tagging: learning

- Chunck parsing -> tagging
- *n* chuncks -> *2n+1* tags
- **B**egin , **I**nside, **O**utside



E se le CFG non ci piacciono?

Back in U.S.S.R.

-> grammatiche a dipendenze